Hamiltonians of mean force

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We study the operator representability problem for the Hamiltonian of mean force (HMF) in Gaussian linearly coupled environments beyond the commuting limit. Building on the quenched-density construction, we derive an exact imaginary-time reformulation in which bath statistics enter as scalar kernel moments while operator growth is generated by the adjoint chain $\{\mathrm{ad}^n_{H_Q}(f)\}_{n\geq 0}$. This yields a structural criterion: a finite closed-form HMF exists only when the adjoint-generated operator algebra closes inside the target ansatz. We then instantiate the criterion in a Pauli-closed qubit model and obtain an explicit closed-form mean-force Hamiltonian with temperature- and bath-dependence carried by mode-resolved kernel coefficients. The manuscript is framed as the noncommuting Gaussian follow-up to the commuting benchmark analysis of arXiv:2602.13146 and sets up the non-Gaussian extension program.

INTRODUCTION

This manuscript is a direct follow-up to Ref. [1] (arXiv:2602.13146). That paper established the quenched-density representation and delivered an exact benchmark in the commuting Gaussian sector. The present paper starts from that benchmark and addresses the next step: how the Hamiltonian of mean force is organised once the coupling operator does *not* commute with the bare system Hamiltonian.

In closed quantum statistical mechanics, equilibrium is generated by a Hamiltonian: $\rho \propto e^{-\beta H}$. For an open system with finite coupling, the operationally defined equilibrium state of the subsystem is the reduced state of the global Gibbs ensemble,

$$\bar{\rho}_Q(\beta) = \mathrm{Tr}_X\, e^{-\beta H_{\mathrm{tot}}}, \tag{1}$$

$$e^{-\beta H_{\mathrm{MF}}(\beta)} \propto \frac{\mathrm{Tr}_X e^{-\beta H_{\mathrm{tot}}}}{Z_X(\beta)}, \quad Z_X(\beta) = \mathrm{Tr}_X e^{-\beta H_X}. \tag{2}$$

This object is generally not $e^{-\beta H_Q}$ for the bare system Hamiltonian H_Q. The resulting representational question is precise: what operator, if any, plays the role of an equilibrium generator for the subsystem once the coupling is non-negligible? The Hamiltonian of mean force (HMF) answers this by construction and is the standard starting point in strong-coupling thermodynamics and the mean-force Gibbs-state program [2–7].

Why care about a mean-force Hamiltonian at all? First, it provides the exact reduced equilibrium object that underlies strong-coupling thermodynamic identities, including free-energy and work relations that remain valid beyond weak coupling[2, 5, 8–11]. Second, it furnishes a consistent equilibrium initialization for open-system dynamics when correlations with the bath are unavoidable, a point emphasized in the literature on correlated initial states and reduced dynamics [4, 12]. Related subensemble approaches explicitly treat the subsystem thermodynamics as inherited from a global canonical ensemble, making the effective reduced generator central to the formalism[13]. In short, the HMF is not an optional reinterpretation; it is the exact operator that encodes the reduced equilibrium state whenever system–bath coupling is finite.

Historically, open-system theory prioritized weak-coupling and Markovian regimes, where reduced equilibrium can often be approximated by a Gibbs state of a renormalized H_Q. At finite coupling the reduced state inherits explicit temperature dependence and interaction-induced operator content that is not captured by a simple renormalization. Coupling-dependent thermodynamic response features in quantum Brownian motion and related models highlight this complexity[14, 15]. The strong-coupling literature consequently treats the mean-force Gibbs state as a distinct equilibrium object, with operational ramifications for heat and energy definitions[9, 16–18]. The HMF can encode interaction-induced operator content absent from H_Q [14, 15, 19], making explicit representation of this content a primary goal.

The HMF literature itself is broad but structured. Canonical definitions and thermodynamic identities are developed in strong-coupling thermodynamics and fluctuation-relation work[2, 3, 5, 8]. A comprehensive review consolidates the "static" mean-force Gibbs perspective with the "dynamical" return-to-equilibrium perspective in open quantum systems [4]. Operational questions such as measurability and thermodynamic consistency at strong coupling have also been pursued[9, 17, 18].

Exact or controlled evaluations exist in special cases. For commuting (QND) interactions, the operator algebra closes trivially and the HMF can be written explicitly[20]. Quadratic/Gaussian models (e.g., damped harmonic oscillators) are solvable because Gaussianity is preserved, leading to closed operator forms[21–23]. Finite-dimensional closures such as spin-boson or single-qubit models provide additional controlled benchmarks[24]. Beyond these cases, the mean-force Gibbs state is often accessed through systematic limits: Cresser and Anders derive weak- and ultrastrong-coupling expressions and

show that, in the ultrastrong limit, the mean-force Gibbs state becomes diagonal in the interaction basis rather than the system Hamiltonian basis[25, 26]. Recent work generalizes the HMF framework to finite baths by introducing a pair of quantum Hamiltonians of mean force that incorporate bath feedback[27], and structural studies further analyze the operator content and potential renormalization of the HMF in extended settings[19, 28].

Outside solvable models, most approaches are perturbative or numerical. Weak-coupling expansions yield controlled but limited series [25, 29]. Reaction-coordinate mappings[30–33] and polaron transforms provide approximate local generators applicable at intermediate coupling. Imaginary-time path-integral methods, stochastic representations, and hierarchical-equations techniques can compute $\bar{\rho}_Q(\beta)$ directly but typically do not yield compact operator forms for $H_{\rm MF}$[34–39]. Stochastic Liouville and partition-free approaches give complementary numerical access without yielding closed-form HMFs [40, 41]. These methods are indispensable for quantitative predictions but leave open the representational question: when does a local or otherwise restricted operator form exist?

The influence-functional formalism is a natural language for this problem. For Gaussian baths with linear coupling it provides an exact route to integrating out bath degrees of freedom[21, 22, 42]. In equilibrium it becomes a Euclidean (imaginary-time) influence functional, usually bilocal in τ, and admits a Hubbard–Stratonovich rewriting as a quenched Gaussian-field average[43–45]. Related path-integral derivations of quantum Langevin dynamics make explicit the noise and dissipation structure inherited from the bath[46, 47]. This formulation also clarifies the terminology: "nonlocal" initially refers to imaginary-time nonlocality of the kernel, whereas the HMF locality question concerns the operator structure on the system Hilbert space.

The existence of $H_{\rm MF}$ is therefore not the issue—it is defined by a logarithm of a traced exponential. The real obstruction is representability: when does $H_{\rm MF}$ admit a closed-form expression within a restricted operator family (few-body, spatially local, or a given algebra)? Much of the literature either (i) solves special models, (ii) expands in controlled limits, or (iii) computes $\bar{\rho}_Q(\beta)$ numerically and analyzes its properties, but a general structural criterion is still lacking [25, 27, 28].

Relative to Ref. [1], the present manuscript has a narrower but deeper scope. We retain the exact Gaussian influence-functional and quenched-field identities, then use adjoint-action and Magnus/BCH organisation [48–50] to study representability in the noncommuting sector. The central result is an explicit closure criterion: a finite closed-form HMF exists only when the operator algebra generated by repeated adjoint action of H_Q on the coupling operators closes inside the target ansatz. We then instantiate this criterion in a Pauli-closed qubit example, where all bath dependence is reduced to scalar kernel moments and the mean-force Hamiltonian is written in exact closed form. Moreover, it is not often one has the opportunity to say something new about a qubit.

QUENCHED REPRESENTATION AND INFLUENCE FUNCTIONAL

We begin by recapitulating the quenched representation introduced in Ref. [1]. To proceed directly from Sec. , we explicit the composite model. We denote the bare system Hamiltonian by H_Q and write

$$H_{\rm tot} = H_Q + H_X + H_{\rm int}, \tag{3}$$

where H_X is the bath Hamiltonian. We assume a factorizable interaction

$$H_{\rm int} = f \otimes B, \tag{4}$$

where f acts on the system and B is a bath operator. We can define the reduced equilibrium operator (up to normalisation) by $\bar{\rho}_Q(\beta) \equiv {\rm Tr}_X e^{-\beta H_{\rm tot}}$. As shown in Ref. [1], this can be represented as a *quenched density*. This is an average over a stochastic propagator, given by:

$$\bar{\rho}_Q(\beta) = \mathbb{E}_\xi\big[U_\xi(\beta)\big], \tag{5}$$

$$U_\xi(\beta) \equiv \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\,\big(H_Q + \xi(\tau) f\big)\right], \tag{6}$$

where $\xi(\tau)$ is a stochastic process whose statistics encode the bath correlations, and $\mathcal{T}_\tau$ denotes time-ordering in imaginary time. Regardless of the precise form of bath and coupling, the reduced density must always be describable in this form.

This result can be understood intuitively by first observing that since the environment influence can only enter through the system coupling f, its influence can be captured by attaching a τ-dependent driving field to f [51, 52]. If this were a single deterministic field however, it would correspond to the bath exerting the *same* back-action history for every microscopic bath configuration. But ${\rm Tr}_X$ averages over many bath microstates in the thermal ensemble, and hence over many back-action histories. In this sense the partial trace is necessarily an average over histories in imaginary time, and the quenched representation simply makes this averaging explicit. For each realisation $\xi(\tau)$ the system evolves under an imaginary-time Hamiltonian $H_Q + \xi(\tau) f$; the bath is then recovered by averaging over $\xi(\tau)$ with a law chosen to reproduce the bath-induced correlations. In this sense $\xi(\tau)$ is not a physical external control field but an efficient parametrisation of the bath history $B(\tau)$ as seen through the coupling channel.

We may understand what formal properties are demanded of ξ by considering the influence functional it is

required to match. Working in imaginary time, introduce the bath interaction picture:

$$B(\tau) \equiv e^{\tau H_X} B e^{-\tau H_X}, \qquad \tau \in [0, \beta], \tag{7}$$

and define the bath thermal state $\rho_X \equiv e^{-\beta H_X}/Z_X$. For an arbitrary c-number source $j(\tau)$ coupled linearly to $B(\tau)$, the bath generates a (time-ordered) functional

$$\mathcal{Z}_X[j] \equiv \mathrm{Tr}_X \left[\mathcal{T}_\tau \exp\left(-\int_0^\beta d\tau\, j(\tau)\, B(\tau) \right) \rho_X \right]. \tag{8}$$

Here $j(\tau)$ is introduced as an external c-number source used to generate ordered bath correlators by functional differentiation. More generally, $j(\tau)$ may be any object commuting with the bath algebra (e.g. a system operator tensored with $\mathbb{I}_X$). In the influence-functional derivation it is ultimately supplied by the system history, which becomes a c-number function in the path-integral representation. In the influence-functional approach, tracing out the bath produces precisely such a functional, evaluated on the system history through the coupling channel.

From this, the bath contribution to the effective Euclidean action can be written as [21, 22, 42]

$$\mathcal{F}[f] \equiv \mathcal{Z}_X[f], \qquad \Phi[f] \equiv \log \mathcal{F}[f]. \tag{9}$$

A key structural fact is that $\Phi[f] = \log \mathcal{F}[f]$ is the *cumulant generating functional* of the bath operator $B(\tau)$ with respect to the thermal state. Concretely, the generalised (time-ordered) cumulant theorem implies the connected expansion [53, 54]

$$\begin{aligned} \Phi[f] = \sum_{n\geq 1} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \cdots d\tau_n \\ \times K^{(n)}(\tau_1, \ldots, \tau_n)\, f(\tau_1) \cdots f(\tau_n), \end{aligned} \tag{10}$$

where the kernels $K^{(n)}$ are the *connected* (cumulant) bath correlators, defined via [55]

$$K^{(n)}(\tau_1, \ldots, \tau_n) \equiv \left\langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \right\rangle_c. \tag{11}$$

The explicit connection back to Eq. (8) is given via its functional differentiation:

$$\left\langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \right\rangle_c = (-1)^n \left. \frac{\delta^n \log \mathcal{Z}_X[j]}{\delta j(\tau_1) \cdots \delta j(\tau_n)} \right|_{j=0}. \tag{12}$$

The bath influence is then completely characterised by the hierarchy $\{K^{(n)}\}_{n\geq 1}$.

To connect the influence functional back to the quenched density, we use the fact that the bath influence depends on the system history only through the linear functional $\int_0^\beta d\tau\, f(\tau) B(\tau)$. One may therefore represent $\mathcal{F}[f]$ as the (generalised) characteristic functional of an auxiliary field $\xi(\tau)$ [43–45]:

$$\mathcal{F}[f] = \mathbb{E}_\xi \left[\exp\left(-\int_0^\beta d\tau\, \xi(\tau)\, f(\tau) \right) \right], \tag{13}$$

where $\mathbb{E}_\xi[\cdot]$ denotes averaging with respect to a (possibly complex) measure on ξ-histories chosen such that Eq. (13) holds.

Since $\xi(\tau)$ is a commuting c-number field, its n-point moments are symmetric under permutations of the time arguments. The influence kernels $K^{(n)}$ fix the *cumulants* of ξ via

$$\left\langle \xi(\tau_1) \cdots \xi(\tau_n) \right\rangle_c = K^{(n)}(\tau_1, \ldots, \tau_n). \tag{14}$$

Consequently, the ordinary correlation functions of the noise are obtained from $\{K^{(m)}\}$ by the standard moment-cumulant relations:

$$\left\langle \xi(\tau_1) \cdots \xi(\tau_n) \right\rangle = \sum_{\pi \in \mathcal{P}_n} \prod_{C \in \pi} K^{(|C|)}\big(\{\tau_i\}_{i\in C}\big), \tag{15}$$

where $\mathcal{P}_n$ denotes the set of all partitions of the index set $\{1, \ldots, n\}$, and C are the disjoint blocks of a given partition $\pi \in \mathcal{P}_n$. After shifting the mean so that $K^{(1)}(\tau) = \langle \xi(\tau) \rangle = 0$, one has (for example)

$$\left\langle \xi(\tau_1)\xi(\tau_2) \right\rangle = K^{(2)}(\tau_1, \tau_2), \tag{16}$$

$$\left\langle \xi(\tau_1)\xi(\tau_2)\xi(\tau_3) \right\rangle = K^{(3)}(\tau_1, \tau_2, \tau_3), \tag{17}$$

$$\begin{aligned} &\left\langle \xi(\tau_1)\xi(\tau_2)\xi(\tau_3)\xi(\tau_4) \right\rangle \\ &\quad = K^{(4)}(\tau_1, \tau_2, \tau_3, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_2) K^{(2)}(\tau_3, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_3) K^{(2)}(\tau_2, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_4) K^{(2)}(\tau_2, \tau_3). \end{aligned} \tag{18}$$

QUENCHED DENSITY AND THE HAMILTONIAN OF MEAN FORCE

The Hamiltonian of mean force is, by definition, the operator whose Gibbs form reproduces the system's reduced equilibrium state. Equivalently, it is the *operator logarithm* of the unnormalised reduced equilibrium operator

$$\bar{\rho}_Q(\beta) \equiv \mathrm{Tr}_X e^{-\beta H_{\mathrm{tot}}}, \qquad H_{\mathrm{MF}}(\beta) \;\equiv\; -\frac{1}{\beta} \log \bar{\rho}_Q(\beta), \tag{19}$$

defined up to an additive multiple of the identity (fixed only when normalising the state).

The quenched representation in Eq. (5) supplies an exact stochastic parametrisation of the unnormalised mean-force Gibbs operator $\bar{\rho}_Q(\beta)$ by making the bath trace an explicit average over imaginary-time back-action histories. Combining Eq. (5) with Eq. (19) yields

$$H_{\mathrm{MF}}(\beta) = -\frac{1}{\beta} \log \mathbb{E}_\xi[U_\xi(\beta)]\,. \tag{20}$$

Thus, constructing the mean-force Hamiltonian reduces to evaluating a stochastic average and then compressing

the result via an operator logarithm. The conditions for being able to perform this average exactly is the focus of the present work.

To make the handling of this problem more concrete, we shall specialise the environment to the Caldeira-Leggett model, where the bath is a collection of harmonic oscillators ($H_X = \sum_k \omega_k b_k^\dagger b_k$) and the coupling is linear in bath coordinates ($B = \sum_k c_k x_k$). None of the results that follow are essentially dependent on this choice, and a generalisation to anharmonic environments is (relatively) straightforward. In the interests of comprehensibility however, we restrict our scope to quadratic environments. In this setting, the auxiliary field $\xi(\tau)$ becomes a stationary zero-mean Gaussian process completely characterized by its covariance

$$\mathbb{E}_\xi[\xi(\tau)\xi(\tau')] = K(\tau - \tau'). \tag{21}$$

The kernel $K(\tau)$ is determined by the bath spectral density $J(\omega) = \frac{\pi}{2}\sum_k \frac{c_k^2}{m_k\omega_k}\delta(\omega - \omega_k)$ via the relation [56]

$$K(\tau) = \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\,\frac{\cosh\big(\omega(\beta/2 - |\tau|)\big)}{\sinh(\beta\omega/2)}. \tag{22}$$

A key property of this kernel is its integrated strength. Integrating Eq. (22) yields

$$\begin{aligned}\int_0^\beta d\tau\, K(\tau) &= \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\int_0^\beta d\tau\,\frac{\cosh\big(\omega(\beta/2 - |\tau|)\big)}{\sinh(\beta\omega/2)}\\ &= \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\,\frac{2}{\omega}\\ &= 2\lambda,\end{aligned} \tag{23}$$

where λ is the explicit reorganisation energy. Consequently, the total variance of the integrated noise field $\Xi = \int_0^\beta d\tau\,\xi(\tau)$ grows linearly with inverse temperature:

$$\mathbb{E}_\xi[\Xi^2] = \int_0^\beta d\tau \int_0^\beta d\tau'\, K(\tau - \tau') = 2\beta\lambda. \tag{24}$$

In the case that the system and its coupling commute, $[H_Q, f] = 0$ and f is τ-independent in imaginary time. In this instance time ordering drops out, and the average is given by [1]:

$$\bar{\rho}_Q(\beta) = \exp\left[-\beta\left(H_Q - \frac{\kappa_0(\beta)}{2}f^2\right)\right], \tag{25}$$

which in turn yields

$$H_{\mathrm{MF}}(\beta) = H_Q - \frac{\kappa_0(\beta)}{2}f^2 + \frac{1}{\beta}\log Z_X(\beta)\,\mathbb{I}. \tag{26}$$

Notably this correction to $H_{\mathrm{MF}}(\beta)$ is entirely *classical* [1]. This is hardly surprising, but emphasises that quantum effects *must* stem from non-commutativity. A truly quantum theory of thermodynamics therefore only makes meaningfully distinct predictions when $[H_Q, f] \neq 0$. In this case however, the noise enters through a noncommuting operator inside $\mathcal{T}_\tau$, rendering the question of averaging highly non-trivial. In the next section, we attack this problem directly, deriving conditions under which $H_{\mathrm{MF}}(\beta)$ possesses a closed form.

GENERATING-FUNCTION STRUCTURE OF THE INFLUENCE OPERATOR

Quenched propagator and Gaussian resummation

The starting point is the quenched propagator introduced in Sec. . In the imaginary-time interaction picture with respect to H_Q, the unnormalised reduced equilibrium operator takes the form

$$\begin{aligned}\bar{\rho}_Q(\beta) &= e^{-\beta H_Q}\,\langle W_\xi(\beta)\rangle_\xi,\\ W_\xi(\beta) &\equiv \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\,\xi(\tau)\,\tilde{f}(\tau)\right],\end{aligned} \tag{27}$$

where $\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q}$ and the noise covariance is $\langle\xi(\tau)\xi(\tau')\rangle_\xi = K(\tau - \tau')$.

Because the bath is Gaussian, only the second cumulant of ξ is non-vanishing. Time-ordering introduces a non-trivial operator structure, but the key identity—proved in Appendix A—is that the Wick resummation of all pairings exponentiates exactly:

$$\langle W_\xi(\beta)\rangle_\xi = \exp\left[\frac{1}{2}\int_0^\beta d\tau\int_0^\beta d\tau'\, K(\tau - \tau')\,\mathcal{T}_\tau\big[\tilde{f}(\tau)\tilde{f}(\tau')\big]\right]. \tag{28}$$

The proof is a direct application of Wick's theorem to the Dyson expansion of W_ξ: at order $2p$ in the expansion, there are $(2p)!/(2^p p!)$ distinct Wick pairings, each weighted by the two-point kernel $K(\tau_i - \tau_j)$. The prefactors from the Dyson expansion precisely cancel the pairing multiplicity, leaving only the singly-connected pairs.

We can exploit the symmetry $K(\tau - \tau') = K(\tau' - \tau)$ to fold the square domain onto the triangular one (see Appendix A), such that the exponent simplifies to

$$\Delta(\beta) \equiv \int_0^\beta d\tau\int_0^\tau d\tau'\, K(\tau - \tau')\,\tilde{f}(\tau)\tilde{f}(\tau'), \tag{29}$$

and the averaged reduced state takes the product form

$$\bar{\rho}_Q(\beta) = e^{-\beta H_Q}\, e^{\Delta(\beta)}. \tag{30}$$

The influence exponent $\Delta(\beta)$ is quadratic in $\tilde{f}$ and depends on the bath exclusively through the two-point kernel K. Here the absence of higher-order bath objects is not an approximation, but an exact consequence of the Gaussian nature of the environment.

The ordered Green function and frequency compression

At this stage, the operator Δ is still formally defined as an integral over the bath degrees of freedom. There is however a particularly satisfying interpretation of this operator in which the effect of the bath is readily understood. To make this explicit, we work in the eigenbasis of H_Q. Let $H_Q|i\rangle = E_i|i\rangle$, so that the matrix elements of the interaction-picture operator are

$$\tilde{f}_{ij}(\tau) = \langle i|\tilde{f}(\tau)|j\rangle = f_{ij}\, e^{\omega_{ij}\tau}, \qquad \omega_{ij} \equiv E_i - E_j. \quad (31)$$

Substituting into Eq. (29) and expanding in the eigenbasis, the influence exponent takes the spectral form

$$\Delta(\beta) = \sum_{i,\ell} \Delta_{i\ell}\, |i\rangle\langle\ell|, \quad (32)$$

where evaluating the matrix elements of the operator product yields

$$\Delta_{i\ell} = \sum_j f_{ij}\, f_{j\ell}\, G^{>}(\omega_{ij}, \omega_{j\ell}), \quad (33)$$

where all bath and temperature dependence has been absorbed into the *ordered Green function*

$$G^{>}(\omega_1, \omega_2) \equiv \int_0^\beta d\tau \int_0^\tau d\tau'\, K(\tau - \tau')\, e^{\omega_1\tau + \omega_2\tau'}. \quad (34)$$

In standard finite-temperature perturbation theory, analogous time-ordered integrals are evaluated using discrete Matsubara frequencies. Here, the integration instead projects the bath response directly onto the Laplace frequencies set by the system's own energy differences; the superscript $>$ records that $\tau > \tau'$ throughout the integration domain.

The double integral in Eq. (34) collapses to a one-dimensional Laplace transform upon changing variables to $u = \tau - \tau'$ and $v = \tau'$:

$$G^{>}(\omega_1, \omega_2) = \frac{e^{\beta\omega_1}\mathcal{K}(\omega_2) - \mathcal{K}(\omega_1)}{\omega_1 + \omega_2}, \quad (35)$$

where

$$\mathcal{K}(\omega) \equiv \int_0^\beta du\, K(u)\, e^{\omega u} \quad (36)$$

is the bare Laplace transform of the bath kernel. Eq. (35) is an exact closed form valid for all $\omega_1 + \omega_2 \neq 0$. Importantly, it implies that for any transition between levels in the reduced system, the entirety of the bath's influence is encoded by $\mathcal{K}$ evaluated at the transition frequency.

The resonant channel $\omega_1 + \omega_2 = 0$ requires separate treatment. Taking the limit carefully yields

$$R(\omega) \equiv G^{>}(\omega, -\omega) = \int_0^\beta (\beta - u)\, K(u)\, e^{\omega u}\, du, \quad (37)$$

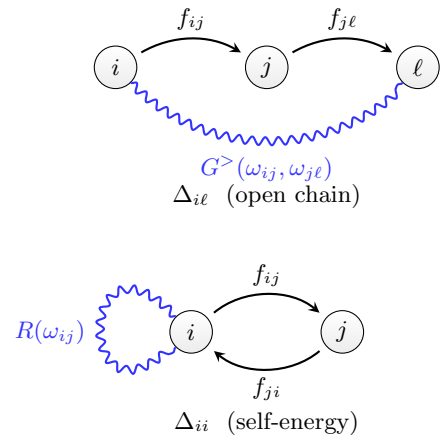

FIG. 1. Diagrammatic structure of the influence exponent $\Delta(\beta)$ in the H_Q eigenbasis. *Top*: The off-diagonal element $\Delta_{i\ell}$ (open chain) sums intermediate sites j, weighted by the bath propagator $G^{>}(\omega_{ij}, \omega_{j\ell})$ (wavy line) and two coupling insertions f (arrows). Energy is not conserved at the intermediate vertex. *Bottom*: The diagonal element Δ_{ii} (self-energy) is a closed loop $i \to j \to i$; the resonant condition $\omega_1 + \omega_2 = 0$ is satisfied, and the bath propagator collapses to $R(\omega_{ij})$.

where the factor $(\beta - u)$ is the available imaginary time after a bath correlation at separation u; the resonant kernel thus accumulates bath memory over the full thermal interval, regulated by the finiteness of β.

The structure of Eq. (33) admits a pleasingly simple diagrammatic reading, illustrated in Fig. 1. To isolate the physically distinct effects of the bath, we partition the influence exponent into its secular (diagonal) and mixing (off-diagonal) components, $\Delta = \Delta_\parallel + \Delta_\perp$.

The secular sector $\Delta_\parallel \equiv \sum_i \Delta_{ii}|i\rangle\langle i|$ singles out the *closed loops* $i \to j \to i$, for which $\omega_{ij} + \omega_{ji} = 0$ identically. The resonance condition forces the bath propagator onto $R(\omega_{ij})$, the object that accumulates bath memory over the whole thermal interval. The diagonal elements are thus the exact, non-perturbative *self-energy* dressing of the i-th energy level by all levels j it is coupled to:

$$\Delta_{ii} = \sum_j |f_{ij}|^2\, R(\omega_{ij}). \quad (38)$$

Conversely, the mixing sector $\Delta_\perp \equiv \sum_{i\neq\ell} \Delta_{i\ell}|i\rangle\langle\ell|$ describes *open chains* $i \to j \to \ell$, where energy is not conserved at the intermediate vertex. Using the closed form in Eq. (35), these components can be expressed directly in terms of the Laplace-transformed bath kernel $\mathcal{K}(\omega)$:

$$\Delta_{i\ell} = \frac{1}{\omega_{i\ell}} \sum_j f_{ij} f_{j\ell} \left[e^{\beta\omega_{ij}}\mathcal{K}(\omega_{j\ell}) - \mathcal{K}(\omega_{ij}) \right] \quad (i \neq \ell), \quad (39)$$

which encodes the coherent transitions between distinct energy levels mediated by the asymmetric fluctuations of the bath.

KMS symmetry and the symmetrised reduced state

The bath kernel $K(\tau)$ satisfies the KMS periodicity condition $K(\tau) = K(\beta-\tau)$, which in Laplace space reads

$$\mathcal{K}(-\omega) = e^{-\beta\omega}\,\mathcal{K}(\omega). \tag{40}$$

Substituting into the closed form Eq. (35) and simplifying gives a corresponding symmetry of the ordered Green function:

$$G^{>}(-\omega_2,-\omega_1) = e^{-\beta(\omega_1+\omega_2)}\,G^{>}(\omega_1,\omega_2). \tag{41}$$

Because transition frequencies telescope, $\omega_{ij}+\omega_{j\ell}=\omega_{i\ell}$, applying Eq. (41) to the matrix-element formula Eq. (33) immediately yields the *detailed-balance* constraint

$$\Delta_{\ell i} = e^{-\beta\omega_{i\ell}}\,\Delta^*_{i\ell}, \tag{42}$$

or in operator form $\Delta^\dagger = \Pi\,\Delta\,\Pi^{-1}$ where $\Pi \equiv e^{-\beta H_Q}$. Although Δ itself is not self-adjoint, this relation is a precise constraint: the product $\Pi\Delta$ is Hermitian.

This symmetry allows us to recast the unnormalised reduced state Eq. (30) into a manifestly positive, Hermitian form. Writing the bare Gibbs factor as $\Pi \equiv e^{-\beta H_Q}$ gives $\bar\rho_Q = \Pi\,e^{\Delta}$. By splitting $\Pi = \Pi^{1/2}\Pi^{1/2}$ and inserting the identity $\Pi^{-1/2}\Pi^{1/2} = \mathbf{1}$, we symmetrically distribute the bare weight:

$$\bar\rho_Q = \Pi^{1/2}\big(\Pi^{1/2}\,e^{\Delta}\,\Pi^{-1/2}\big)\Pi^{1/2}. \tag{43}$$

Applying the operator identity $A\,e^{B}A^{-1} = \exp(ABA^{-1})$ to the inner bracket yields

$$\bar\rho_Q = \Pi^{1/2}\,e^{S}\,\Pi^{1/2}, \qquad S \;\equiv\; \Pi^{1/2}\,\Delta\,\Pi^{-1/2}, \tag{44}$$

where S is the *symmetrised influence operator*. Its matrix elements evaluate to $S_{i\ell} = e^{-\beta\omega_{i\ell}/2}\Delta_{i\ell}$, which by Eq. (42) satisfy $S^*_{i\ell} = S_{\ell i}$. Thus, the modified exponent S is strictly self-adjoint. The partition function of the reduced system follows by cyclicity of the trace:

$$Z_Q \;\equiv\; \mathrm{Tr}[\bar\rho_Q] = \mathrm{Tr}[\Pi\,e^{S}] = \sum_i e^{-\beta E_i}\,(e^{S})_{ii}. \tag{45}$$

Since S is self-adjoint, $(e^S)_{ii} \geq e^{S_{ii}} = e^{\Delta_{ii}}$ by Jensen's inequality applied to the spectral weights $|\langle i|n\rangle|^2$, giving the bound

$$Z_Q \;\geq\; \sum_i e^{-\beta E_i+\Delta_{ii}}, \tag{46}$$

where the right-hand side is a Boltzmann sum with bath-shifted energies and equality holds when S is diagonal (i.e. when H_Q and Δ share an eigenbasis). The full result $Z_Q = \mathrm{Tr}[\Pi e^S]$ captures all level-mixing corrections through the off-diagonal elements of S.

THE HAMILTONIAN OF MEAN FORCE

Closure as an operator-algebra statement

With this, we may now give the general expression for the Hamiltonian of Mean-Force.

$$-\beta H_{\mathrm{MF}}(\beta) = \log\big(e^{-\beta H_Q}\,e^{\Delta(\beta)}\big). \tag{47}$$

If H_Q and Δ commuted, the logarithm would trivially give $-\beta H_Q + \Delta$. In general they do not, and the Baker–Campbell–Hausdorff (BCH) expansion

$$\log(e^{A}e^{B}) = A+\Phi(\mathrm{ad}_A)\,B+O(B^2), \qquad \Phi(x) \equiv \frac{x}{1-e^{-x}}, \tag{48}$$

generates a tower of commutators of $A = -\beta H_Q$ acting on $B = \Delta$. To understand the structure of these corrections, we invoke the partition of the influence exponent into its secular and mixing parts, $\Delta = \Delta_\parallel + \Delta_\perp$. The structural partitioning of these mean-force corrections is illustrated in Fig. 2. Because the secular piece $\Delta_\parallel$ strictly commutes with A, its adjoint action is annihilated ($\mathrm{ad}_A(\Delta_\parallel) = 0$).

Consequently, the generating function evaluated at $x = 0$ yields $\Phi(0) = 1$, allowing the parallel component to push straight through the linear BCH expansion:

$$-\beta H_{\mathrm{MF}} \approx \big(-\beta H_Q + \Delta_\parallel\big) + \Phi(\mathrm{ad}_{-\beta H_Q})\,\Delta_\perp + O(\Delta^2). \tag{49}$$

The secular zeroth-order shift $\Delta_\parallel$ thus groups directly with the bare system (reminiscent of the weak-coupling Lamb shift). Crucially however, while it acts geometrically like a classical energy shift, this term is still fundamentally *quantum* in origin because it is weighted by the bath Green functions, dynamically entangling the bare spectrum with the full spectral density of the environment. We can understand this directly by reference to Eq. (38). By defining a bath-dressed coupling operator $\tilde f$ with matrix elements $\tilde f_{ij} = f_{ij}\sqrt{R(\omega_{ij})}$, the secular exponent takes the exact quadratic form $\Delta_\parallel = (\tilde f\tilde f^\dagger)_{\mathrm{diag}}$. In the macroscopic limit where the system commutes with its coupling ($[H_Q, f] = 0$), the matrix f is entirely diagonal in the energy basis ($\omega_{ij} = 0$), and this dressed square collapses precisely to $\Delta_\parallel = \beta\lambda f^2$, where $R(0) = \beta\lambda$ is the explicit reorganisation energy defined in Eq. (23). The zeroth order BCH term therefore exactly recovers the classical commuting shift $H_{\mathrm{MF}} = H_Q - \lambda f^2$ of Eq. (26), and at zeroth order the non-commuting correction can be obtained by replacing $f \to \tilde f$. The remaining coherence-generating effects are then driven exclusively by the action of Φ on the perpendicular, off-diagonal open-chain fragments $\Delta_\perp$.

There is a saitsfyingly clean expression for these non-commuting corrections in the system energy eigenbasis. The off-diagonal basis operators $|i\rangle\langle j|$ are exact eigenoperators of the adjoint map: $\mathrm{ad}_{-\beta H_Q}|i\rangle\langle j| =$

$-\beta[H_Q, |i\rangle\langle j|] = -\beta\omega_{ij}|i\rangle\langle j|$. Consequently, the entire infinite tower of nested commutators generated by $\Phi(\mathrm{ad}_{-\beta H_Q})$ trivially evaluates to a simple scalar multiplication for each matrix element of the perpendicular fragment:

$$\Phi(\mathrm{ad}_{-\beta H_Q})\Delta_\perp = \sum_{i\neq j} \Phi(-\beta\omega_{ij})\,\Delta_{ij}|i\rangle\langle j|, \tag{50}$$

where $\Phi(x) = x/(1-e^{-x})$. The off-diagonal elements of the linear-in-Δ effective Hamiltonian are therefore simply the bare bath-induced transitions Δ_{ij}, parametrically reweighted by the thermal asymmetry factor $\Phi(-\beta\omega_{ij}) = -\beta\omega_{ij}/(1-e^{\beta\omega_{ij}})$.

This scalar evaluation exhibits a manifestly physical symmetry. If we reconstruct the effective matrix element using the symmetrised influence operator $S_{ij} = e^{-\beta\omega_{ij}/2}\Delta_{ij}$, the thermal asymmetry factor exactly absorbs the detailed-balance twist. The (off-diagonal) components on the effective Hamiltonian are therefore simply the bare bath-induced transitions Δ_{ij}, parametrically reweighted by the thermal asymmetry factor $\Phi(-\beta\omega_{ij}) = -\beta\omega_{ij}/(1-e^{\beta\omega_{ij}})$:

$$\begin{aligned}(H_{\mathrm{MF}})_{ij} &= -\frac{1}{\beta}\Phi(-\beta\omega_{ij})e^{\beta\omega_{ij}/2}S_{ij} \\ &= -\frac{\omega_{ij}/2}{\sinh(\beta\omega_{ij}/2)}\,e^{-\beta\omega_{ij}/2}\Delta_{ij}.\end{aligned} \tag{51}$$

Because $x/\sinh(x)$ is an explicitly even geometric function of the energy gap ω_{ij}, Eq. (51) guarantees that the linearly truncated effective Hamiltonian is perfectly Hermitian. Moreover, this exact thermal weighting reveals that bath-induced coherent mixing between levels is heavily exponentially suppressed for high-frequency transitions ($|\beta\omega_{ij}| \gg 1$), dynamically localising the leading mean force corrections to the thermodynamic energy shell.

This fortuitous cancellation reveals a deep structural connection between quantum thermodynamics and Lie algebra. The fundamental detailed balance condition of the bath imposes a thermal asymmetry $e^{-\beta\omega}$ on the bare transition amplitudes. Conversely, the BCH expansion—a purely abstract property of non-commuting logarithms—generates the parametric weighting $\Phi(-\beta\omega) \propto (1-e^{\beta\omega})^{-1}$. The fact that the Lie-algebraic geometry of the thermal exponential perfectly mirrors - and thereby absorbs - the thermodynamic detailed-balance twist of the environment is striking, and illustrative of the high degree of mathematical harmonisation underlying the mean-force formulation.

The foregoing parametric logic naturally generalises directly to all higher orders of the BCH expansion. Because the basis operators $|i\rangle\langle j|$ are strict eigenoperators of the adjoint map, the evaluation of any highly nested Lie commutator string spanning $O(\Delta^n)$ simply produces ordinary matrix products of the abstract fragments $\Delta_{ik_1}\Delta_{k_1k_2}\ldots\Delta_{k_{n-1}j}$, completely bypassing the need for functional derivatives. Over the energy eigenbasis, the entire non-linear algebraic sum reduces to evaluating n-step discrete pathways through the bare spectrum, where each composite transition amplitude is identically parametrically weighted by a universal, symmetric multivariate generalisation of Φ determined solely by Lie combinatorics.

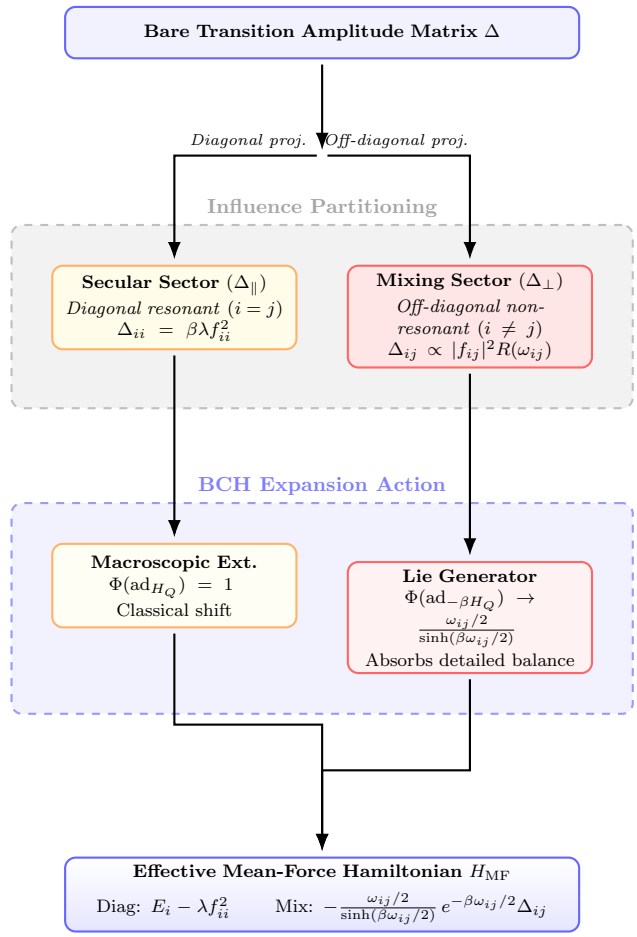

FIG. 2. Structural decomposition of the linear-order Mean-Force Hamiltonian. The parallel (secular) sector generates classical Lamb-like commuting phase shifts, entirely isolated from detailed balance effects. Concurrently, the perpendicular (mixing) sector passes through the Baker–Campbell–Hausdorff non-commutative generator, producing a geometric Lie scaling that perfectly absorbs the detailed balance thermal twist.

While this geometric pathing approach is exceptionally suggestive—and likely a numerically robust framework for resolving perturbative components—the abstract operator structure is more directly revealing of the core algebraic symmetries. Because the adjoint action obeys the rule $\mathrm{ad}_{H_Q}(f_n f_m) = f_{n+1}f_m + f_n f_{m+1}$ (where $f_n \equiv \mathrm{ad}_{H_Q}^n f$), every nested BCH commutator stays within the algebra generated by the adjoint chain of the coupling.

This ensures that even high-order nonlinear corrections introduce *no new bath objects*: the environment continues to enter exclusively through the fundamental spectral functions $\mathcal{K}(\omega)$ and $R(\omega)$, with coefficients determined solely by universal Lie combinatorics.

The closure content of $H_{\mathrm{MF}}(\beta)$ is therefore the following: the entire operator lives in the finite-dimensional algebra generated by $\{H_Q, \Delta\}$, with Δ quadratic in f and the bath entering exclusively through the two spectral objects $\mathcal{K}(\omega)$ and $R(\omega)$. Whether this algebra is itself finite-dimensional - and hence whether H_{MF} admits a closed form in a restricted operator family - depends on the structure of f and H_Q This statement leads to an important corollary. For continuous variable systems $H_Q = p^2/2m + V(x)$, if the potential $V(x)$ is a polynomial of degree greater than 2, the adjoint chain $\{f_n\}$ generally does not close in any finite-dimensional Lie algebra; instead, it generates the full universal enveloping algebra. Consequently, no finite representation of the mean force exists within the original operator class. This obstruction is analogous (in spirit) to the Groenewold–Van Hove no-go theorem on consistent quantisation of polynomial observables: the failure to close the Lie algebra under adjoint action corresponds to the impossibility of mapping the quantum evolution of non-quadratic potentials to a finite phase space flow. For such potentials, there is no *local operator* that exactly describes the reduced equilibrium state.

From a practical standpoint, the closure of the adjoint-chain is not necessary for an arbitrarily accurate approximation of $H_{\mathrm{MF}}(\beta)$, as the BCH logarithm $\log(e^{-\beta H_Q} e^{\Delta})$ still yields a controlled expansion. From this algebraic perspective, the closure problem offers a unifying view of strong-coupling approximations. Numerical schemes such as the polaron transformation [30, 56], hierarchical equations of motion [37], and chain-mapping/DMRG techniques [39] can be understood as distinct choices of a truncated operator family $\mathcal{A}$. The polaron frame corresponds to dressing $\mathcal{A}$ with coherent displacements, HEOM truncates the memory depth of the influence functional (dual to the operator degree in the adjoint chain), and chain mappings truncate the spatial extent of the harmonic bath. The 'closure' problem is thus equivalent to identifying a low-dimensional subalgebra that approximately contains the logarithm of the quenched propagator.

EXACT SOLUTION: GEOMETRIC REPRESENTATION

We now put the machinery of Secs. – to work in the transverse-coupling spin-boson model. The qubit is the prototypical rank-1 case of the general framework, yet its $\mathfrak{su}(2)$ algebra is rich enough to generate all the essential non-perturbative physics. As illustrated in Fig. 3, the exact qubit mean-force state $\mathbf{r}$ lives in a three-dimensional Bloch sphere, but is geometrically locked to the *coupling plane* (shaded) spanned by the system axis $\hat{\mathbf{n}}_s$ and the interaction axis $\hat{\mathbf{f}}$. The orientation of this plane is defined by the interaction phase ϕ, which directly determines the phase of the induced coherences in the reduced state.

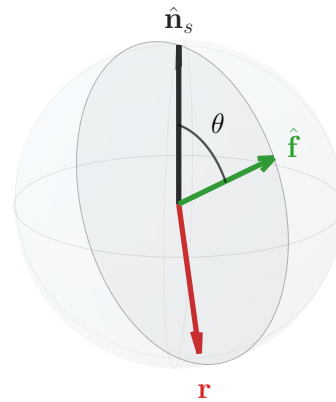

FIG. 3. 3D geometric overview of the exact qubit mean-force state. The system is defined by the bare axis $\hat{\mathbf{n}}_s$ and the interaction axis $\hat{\mathbf{f}}$ at polar angle θ and azimuthal angle ϕ. The exact reduced-state Bloch vector $\mathbf{r}$ is constrained to the shaded coupling plane. The interaction phase ϕ rotates the coupling plane and sets the phase of the steady-state coherences.

We first define our system Hamiltonian and coupling via

$$H_Q = \frac{\omega_q}{2} \hat{\mathbf{n}}_s \cdot \boldsymbol{\sigma}, \qquad f = \hat{\mathbf{f}} \cdot \boldsymbol{\sigma}, \tag{52}$$

where $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{f}}$ are unit vectors defining the system and coupling axes, respectively. The overall coupling strength g is omitted in the following derivation for clarity and can be re-introduced as a global scaling factor where needed. For the derivation it is convenient to work in the eigenbasis of H_Q (so $\hat{\mathbf{n}}_s = \hat{\mathbf{z}}$). The final formulas are then rewritten covariantly in terms of $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{f}}_\perp$.

The influence exponent may easily be calculated directly, using $\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q}$. In the qubit energy basis ($\hat{\mathbf{n}}_s = \hat{\mathbf{z}}$), we decompose the unit coupling vector as $\hat{\mathbf{f}} = (s\cos\phi, s\sin\phi, c)$, where $c \equiv \cos\theta$ and $s \equiv \sin\theta$ denote the longitudinal and transverse projections relative to the system axis, and ϕ is the azimuthal angle. Writing

$$\begin{aligned} f &= c\sigma_z + f_-\sigma_+ + f_+\sigma_-, \\ f_\pm &\equiv s e^{\pm i\phi}, \end{aligned} \tag{53}$$

gives

$$\tilde{f}(\tau) = c\sigma_z + f_- e^{\omega_q \tau}\sigma_+ + f_+ e^{-\omega_q \tau}\sigma_-, \tag{54}$$

meaning $\Delta(\beta)$ is determined solely by the bath response kernels at the frequencies $0, \pm\omega_q$. Consequently, all bath dependence in Δ is compressed to scalar kernel data evaluated at the same frequencies. Using the notation of Sec. , the required bath objects are $\mathcal{K}(0)$, $\mathcal{K}(\pm\omega_q)$, and $R(0)$, $R(\pm\omega_q)$.

For a single qubit, the operator algebra closes almost trivially. Using $\sigma_+\sigma_- = (\mathbb{I} + \sigma_z)/2$, $\sigma_-\sigma_+ = (\mathbb{I} - \sigma_z)/2$, and $\sigma_\pm^2 = 0$, one finds

$$\begin{aligned}\tilde{f}(\tau)\tilde{f}(\tau') = &\Big[c^2 + s^2\cosh\big(\omega_q(\tau-\tau')\big)\Big]\mathbb{I} \\ &+ s^2\sinh\big(\omega_q(\tau-\tau')\big)\sigma_z \\ &+ cf_-\left(e^{\omega_q\tau'} - e^{\omega_q\tau}\right)\sigma_+ \\ &+ cf_+\left(e^{-\omega_q\tau} - e^{-\omega_q\tau'}\right)\sigma_-.\end{aligned} \tag{55}$$

Using the response kernels, the entirety of the bath's influence is compressed into two real, temperature-dependent response kernels: a longitudinal (resonant) kernel Σ_z and a transverse (non-resonant) kernel $\Sigma_\perp$, defined as

$$\begin{aligned}\Sigma_z &\equiv \frac{1}{2}\left[R(\omega_q) - R(-\omega_q)\right], \\ \Sigma_\perp &\equiv \frac{4}{\omega_q}\left[\cosh\left(\frac{\beta\omega_q}{2}\right)\mathcal{K}(0) - e^{-\beta\omega_q/2}\mathcal{K}(\omega_q)\right].\end{aligned} \tag{56}$$

The reduced system's $\mathfrak{su}(2)$ algebra means both the influence exponent and the final reduced state may be calculated exactly. After some algebraic gymnastics (done without harness in App C), the exact reduced density matrix ρ_Q takes the form:

$$\rho_Q = \frac{1}{2}\begin{pmatrix}1 + m_z & m_\perp e^{-i\phi} \\ m_\perp e^{i\phi} & 1 - m_z\end{pmatrix}. \tag{57}$$

Both the definition and the interpretation of the Bloch vector component magnetisations $(m_z, m_\perp)$ relies on several intermediate parametrisations. First and most importantly, we have the *influence magnitude*

$$\chi = s\sqrt{c^2\Sigma_\perp^2 + s^2\Sigma_z^2}, \tag{58}$$

which - as we will see shortly - acts as the essential determinant for the different regimes of behaviour for the system. It is the *norm* of the influence exponent $\|\Delta(\beta)\|^2 \equiv \chi^2$. The reduced state's Bloch vectorradius r is then given by

$$r = \tanh\chi = \sqrt{m_z^2 + m_\perp^2}. \tag{59}$$

Finally, if we further define the factor γ

$$\gamma = \tanh\chi/\chi, \tag{60}$$

then it is possible to write the Bloch vector components as

$$\begin{aligned}m_z &= \tanh\Theta, \\ m_\perp &= \tan\eta\left(m_z\cosh\frac{\beta\omega_q}{2} + \sinh\frac{\beta\omega_q}{2}\right),\end{aligned} \tag{61}$$

with the controlling parameters being given through γ as

$$\begin{aligned}\Theta &\equiv \operatorname{arctanh}(\gamma s^2\Sigma_z) - \frac{\beta\omega_q}{2}, \\ \tan\eta &\equiv -\cot\theta\frac{\Sigma_\perp}{\Sigma_z}.\end{aligned} \tag{62}$$

These final quantities are directly interpretable. Θ is the renormalised effective temperature for the reduced system, and $\tan\eta$ captures the anisotropy of the bath's influence in its longitudinal and transverse components. Notably in the special case of purely transverse coupling $\theta = \frac{\pi}{2}$, the reduced system is simply a renormalised copy of the bare system with $\frac{\beta\omega_q}{2} \to \Theta$.

Since the reduced state is confined to the common plane of $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{f}}$, it is most useful to describe it in terms of its radius r, and its *tilt-angle* φ away from the bare system's axis $\hat{\mathbf{n}}_s$. We write this explicitly in terms of the mixing angle θ as

$$\tan\varphi = -\frac{m_\perp}{m_z} = \frac{\cot\theta\Sigma_\perp}{\cosh\frac{\beta\omega_q}{2}\Sigma_z - \frac{\sinh\frac{\beta\omega_q}{2}}{\gamma s^2}} \tag{63}$$

PHYSICAL REGIMES OF THE MEAN-FORCE STATE

Even in this simple case, the physics encoded in the mean force state by its dependence on temperature and bath coupling is startlingly rich. To expose this, we first commit to a specific bath spectral density for numerics. We choose the Ohmic spectral density

$$J(\omega) = \alpha\,\omega\,e^{-\omega/\omega_c}, \qquad \omega \in [0, 2\omega_q], \tag{64}$$

with $\alpha = 1$ and $\omega_c = 5\omega_q$. Unless otherwise stated, we further specialise to $\theta = \pi/4$ throughout. We emphasise here that the choice of spectral shape is not fundamental, as it only controls the way in which χ runs with temperature.

To expose the coupling dependence cleanly, we reintroduce an explicit dimensionless strength g through $K(u) = g^2K_0(u)$, so that all response kernels scale as g^2. The entire non-perturbative recombination of bare Hamiltonian and influence is then governed by the single dimensionless parameter

$$\chi(\beta, g, \theta) = g^2\,\chi_0(\beta, \theta), \tag{65}$$

through the crossover function

$$\gamma(\chi) = \frac{\tanh\chi}{\chi}. \tag{66}$$

This apparently innocuous expression encodes the entire non-perturbative recombination of bare Hamiltonian and influence, and functions as an *order* parameter for the crossover between weak and strong coupling regimes. The reduced equilibrium state transitions between these regimes as χ passes through unity,

$$\chi(\beta, g, \theta) \sim 1 \qquad \Longleftrightarrow \qquad g^2 \sim \chi_0(\beta, \theta)^{-1}. \tag{67}$$

This crossover condition $\chi(\beta, g, \theta) = 1$ therefore defines a coupling-dependent temperature scale

$$g_\star(\beta) = \chi_0(\beta, \theta)^{-1/2}, \tag{68}$$

and an equivalent crossover temperature $\beta_\star(g)$. Notably, these dual crossover values are an exact property of the solution.

The function $\gamma(\chi)$ is elementary but worth dwelling on. It is monotone decreasing from $\gamma(0) = 1$ to $\gamma(\infty) = 0$ with its inflection near $\chi \approx 1.2$, and admits clean asymptotic forms:

$$\chi \ll 1: \quad \gamma \simeq 1 - \tfrac{\chi^2}{3} \quad \text{(weak coupling)}, \tag{69}$$

$$\chi \gg 1: \quad \gamma \simeq \chi^{-1} = (g^2\chi_0)^{-1} \quad \text{(ultrastrong)}. \tag{70}$$

In the weak-coupling regime the state tracks the bare Gibbs thermal state with $O(g^2)$ corrections, as expected. Notably, as shown in Fig. 4, the crossover between regimes depends on *both* temperature and coupling strength. This means that two distinct regimes emerge in the limit of scaling g and β, wholly dependent on which side of the crossover is taken.

This involutive relationship between temperature and coupling is explored in Fig. 5, which shows how the crossover $\chi = 1$ defines a sharp partition in the (β, g) plane. This line represents a coupling-dependent temperature scale $\beta_\star(g)$ (or equivalently $g_\star(\beta)$), where different parameter scaling relationships apply. A far more nuanced picture emerges in the limit of $\chi \gg 1$, which we defer to Sec. .

State purity

The purity of the qubit mean-force state is $\mathcal{P} = \frac{1}{2}(1 + r^2)$ with $r^2 = |\mathbf{r}|^2$. From this we can immediately establish a universal result:

Bath always purifies. *For any positive spectral density $J(\omega) \geq 0$ and any $\beta > 0$: (i) $r^2(g, \beta)$ is strictly increasing in g for $g \geq 0$, and (ii) for all $g > 0$,*

$$r^2(g, \beta) \;>\; r^2_{\text{bare}} \;\equiv\; \tanh^2\!\left(\frac{\beta\omega_q}{2}\right), \tag{71}$$

with equality only at $g = 0$.

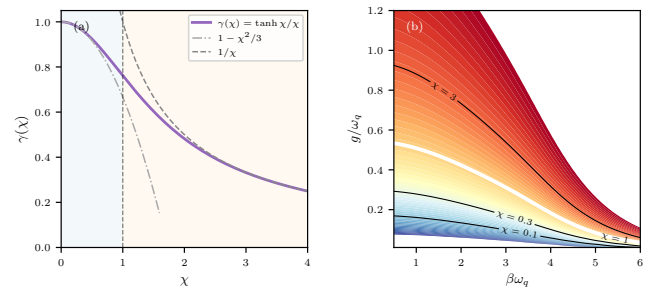

FIG. 4. Universal crossover for the qubit mean-force state. **(a)** Crossover function $\gamma(\chi) = \tanh\chi/\chi$ interpolating between the weak-coupling approximant $1-\chi^2/3$ and the ultrastrong form $1/\chi$. Shaded regions mark the weak (blue) and strong (orange) regimes separated by $\chi = 1$. **(b)** Dimensionless coupling $\chi(\beta, g) = g^2\chi_0(\beta)$ as a contour map. Vertical trajectories (fixed g) will crossover at its corresponding $\beta_\star(g)$.

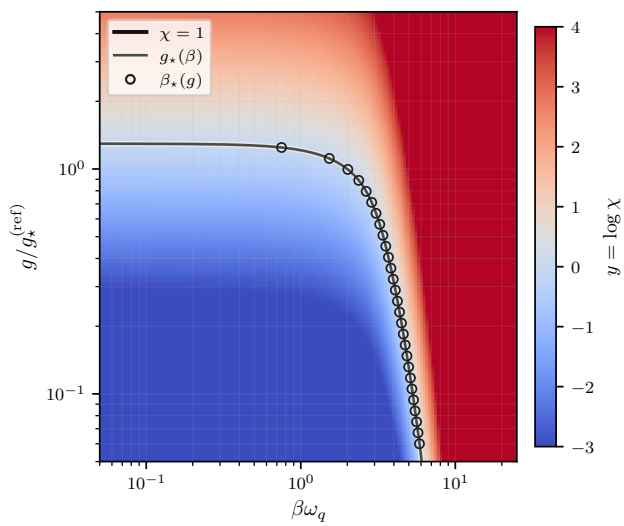

FIG. 5. Involutive crossover geometry generated by $\chi(\beta, g) = g^2\chi_0(\beta)$. The heatmap shows $y = \log\chi$ in the (β, g) plane. The solid black line marks the crossover locus $\chi = 1$, which can be constructed either as a coupling-dependent temperature $\beta_\star(g)$ (circles) or a temperature-dependent coupling $g_\star(\beta)$ (solid grey line). This involutive relationship defines a partition between the weak-coupling (blue) and strong-coupling (red) scaling regimes.

Proof sketch. Part (i) (monotonicity) follows immediately from the exact formula $r = \tanh\chi = \tanh(g^2\chi_0(\beta))$: since tanh is strictly increasing and $g^2\chi_0$ is strictly increasing in g, so is r.

Part (ii) (excess over bare) uses the non-negativity of the resonant (longitudinal) dressing Σ_z. From the explicit integral form of the response kernels (cf. App. C), one has

$$\Sigma_z = \frac{1}{\pi}\int_0^\beta u(\beta - u)\, K(u)\, \frac{\sinh(\omega_q u)}{u}\, du. \tag{72}$$

Since $K(u) \geq 0$ for a physical spectral density, and the remaining factors are non-negative on the integration domain, we have $\Sigma_z > 0$ for all $\beta > 0$.

Substituting the exact components $m_z = \tanh\Theta$ and $m_\perp$ from Eq. (61), the purity excess $\Delta r^2 = r^2 - \tanh^2(\beta\omega_q/2)$ is found to be strictly positive whenever $\gamma s^2 \Sigma_z > 0$. Physically, this reflects the fact that the bath-induced longitudinal field always reinforces the system's tendency to polarise, effectively lowering the local temperature and thus increasing the purity of the state beyond its bare thermal value. □

In the limit $\beta \to 0$ (infinite temperature), the bare state is maximally mixed, yet the mean-force state has

$$r\big|_{\beta\omega_q \to 0} = \tanh\chi > 0, \tag{73}$$

This result is independent of the bandwidth or spectral shape of $J(\omega)$: widening the bath increases $\chi_0(\beta)$ (driving the crossover to weaker g) but cannot reverse the purity inequality. Purity inversion would require reversing the sign of the longitudinal response ($\Sigma_z < 0$), which is impossible for any physical (positive) spectral density.

Figure 6 confirms this numerically. Panel (a) shows $r(g/g_\star)$ for four temperatures. Every trajectory lies strictly above its own r_{bare} reference line for all $g > 0$, confirming (71). Panel (b) shows the purity excess $\Delta r^2 = r^2 - \tanh^2(\beta\omega_q/2)$ over the full (g, β) plane. Enhancement peaks at large g and high temperature (upper-right corner), and falls to near zero at low temperature (left edge) where the bath-independent bare state is already nearly pure. The black dashed contour $\chi = 1$ marks the crossover at which purification accelerates steeply. This is an amusingly counterintuitive result - for sufficiently strong coupling, a *hotter* bath is *more* purifiying!

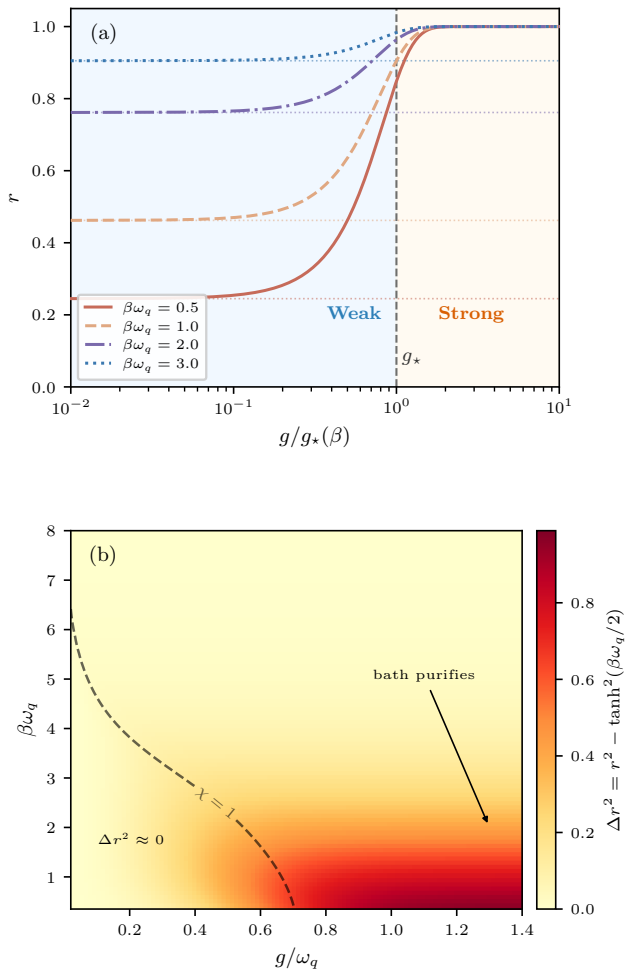

FIG. 6. Purity of the qubit mean-force state. **(a)** Bloch radius r versus scaled coupling $g/g_\star(\beta)$ for four temperatures. Dotted horizontals show the bare thermal purity $r_{\text{bare}} = \tanh(\beta\omega_q/2)$; coupling always increases purity (Eq. (71)), most visibly at high temperature where the bare state is near-maximally-mixed. Shaded regions mark the weak (blue) and strong (orange) coupling regimes. **(b)** Purity excess $\Delta r^2 = r^2 - \tanh^2(\beta\omega_q/2)$ in the (g, β) plane. The black dashed contour marks the crossover locus $\chi = 1$: purification is strongest at large g and high temperature and saturates as $r \to 1$ deep in the strong-coupling regime.

Susceptibility peaks at the crossover

An interesting feature of the crossover regime is that it demarcates the region where the state is maximally sensitive to bath coupling. This is easily seen by considering the susceptibilities of the tilt angle $\partial_g \varphi$ and Bloch radius $\partial_g r$. These quantify how rapidly the Bloch vector tilts and how rapidly the state purifies per unit change in g. As Fig. 7 shows, both peak near the crossover $\chi = 1$. $\partial_g \varphi$ maximises at $g \approx 0.65\, g_\star$ (at $\chi_{\text{peak},\varphi} \approx 0.42$) and $\partial_g r$ maximises at $g \approx 0.92\, g_\star$ ($\chi_{\text{peak,r}} \approx 0.85$). The locations are approximately pinned at universal values of χ independent of temperature, as demonstrated by the alignment of peaks when plotted as a function of the rescaled coupling $g/g_\star(\beta)$.

The crossover region $\chi \sim 1$ is therefore where bath coupling yields the greatest rate of return: both the onset of coherence (tilt of the Bloch vector away from the bare thermal axis) and the onset of purification are most rapid here, before both effects saturate as $\chi \to \infty$. This suggests the crossover regime is the most operationally leveragable for state engineering.

While being most expolitable, this regime is simultaneously the most demanding for numerical simulation. Exact Diagonalisation (ED) of the truncated spin-boson model faces two convergence requirements that become simultaneously binding near $\chi \sim 1$. On the spectral side, a faithful discretisation of $J(\omega)$ requires a mode count N_ω large enough to accurately quadrature the bath kernel at $\omega \in \{0, \pm\omega_q\}$: at the crossover these integrals receive comparable contributions from the full spectral window, so neither the infrared nor the ultraviolet

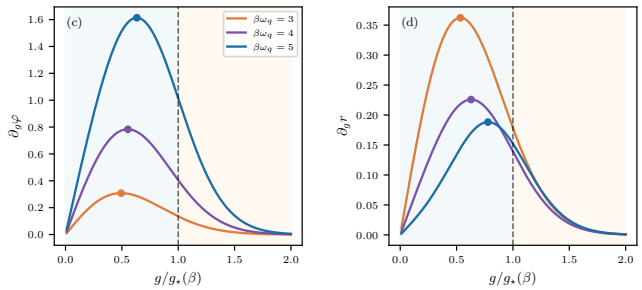

FIG. 7. Coherence and purity susceptibilities (Ohmic bath (64), $\theta = \pi/4$). **(c)** Bloch-angle susceptibility $\partial_g\varphi$ versus scaled coupling $g/g_\star(\beta)$ for three temperatures. Rescaling collapses the peak positions to a universal value $g/g_\star \approx 0.65$, marked by filled circles. **(d)** Purity gradient $\partial_g r$; the rescaled peak occurs at $g/g_\star \approx 0.92$.

modes can be coarsened without systematic bias. On the state-space side, representing the polaron-like mean-force ground state demands a per-mode Fock cutoff $n_{\max} \gtrsim \alpha_k^2$, where the mode displacement $\alpha_k = g\, c_k/\omega_k$ is already appreciable at $g \sim g_\star(\beta)$ yet the state has not saturated to the simple displaced-vacuum structure it acquires only at $\chi \gg 1$. Neither the weak-coupling approximation nor the ultrastrong polaron limit simplifies the representation, so both hierarchies must be converged together. The $\chi = 1$ crossover partitions parameter space into an ED-reliable sector ($\chi \lesssim 1$) and a polaron-representability-limited sector ($\chi \gtrsim 1$), with the susceptibility peaks sitting squarely on this boundary.

Temperature-driven and coupling-driven limits

The parameter $\chi \to \infty$ can be reached along two qualitatively distinct routes through the (g, β) plane, and these routes access different corners of state space.

Coupling-driven limit: $g \to \infty$ at fixed T. As g increases at fixed temperature, the state locks onto the temperature-dependent bath-dressed attractor direction $\hat{\mathbf{h}}_{\text{eff}}(\beta)$. Purity approaches unity, but the direction encodes the temperature: the strong-coupling attractors (squares in Fig. 8(a)) lie at *different* positions for each temperature, reflecting the temperature-dependent weights of the resonant and non-resonant bath channels. This is the *temperature memory* of the mean-force state: even at arbitrarily strong coupling, the equilibrium state carries information about the temperature.

Temperature-driven limit: $T \to 0$ at fixed g. As the system cools at fixed coupling, the bare Gibbs factor dominates and the state approaches the ground state $-\hat{\mathbf{n}}_s$ regardless of g. Panel (b) of Fig. 8 shows this route: curves at fixed g sweep from near the disk centre (high temperature, $\chi \approx 0$) outward to near-unit purity as β grows, converging to $-\hat{\mathbf{n}}_s$ (bottom of the disk) in the limit $T \to 0$. At *large* g the state quickly reaches near-unit purity and then drifts along the disk boundary as the attractor direction shifts with temperature — a precession at approximately constant purity. The endpoint of this precession at $T \to 0$ is always $-\hat{\mathbf{n}}_s$, independent of g.

Co-moving limit. A third natural route holds the scaled coupling $\xi = g/g_\star(\beta)$ fixed as β varies, tracing a level set of $\chi = \xi^2$ in the (g, β) plane. Along such a curve the purity $r = \tanh(\chi) = \tanh(\xi^2)$ is constant, while the Bloch vector direction

$$\hat{\mathbf{n}}(\beta) = \frac{1}{\chi}\big(s\Sigma_\perp c,\ s^2\Sigma_z\big) \tag{74}$$

evolves with temperature. For large ξ (strong coupling), this is an arc at near-unit purity sweeping across the Bloch disk boundary: the strong-coupling attractor precesses as the bath channels $s\Sigma_\perp c/\chi$ and $s^2\Sigma_z/\chi$ vary with β. As $\beta \to 0$ along the fixed-ξ curve, the direction approaches the high-temperature value; as $\beta \to \infty$, the longitudinal channel dominates ($s^2\Sigma_z/\chi \to 1$) and $\hat{\mathbf{n}} \to -\hat{\mathbf{n}}_s$.

The two limits $g \to \infty$ (fixed T) and $T \to 0$ (fixed g) do not commute: the former locks to a temperature-dependent attractor with temperature memory, the latter approaches the bare ground state and forgets the coupling geometry. The co-moving limit interpolates between these extremes at fixed coupling strength.

Relation to the Cresser–Anders ultrastrong limit

It is instructive to make explicit how the exact qubit geometry relates to the ultrastrong mean-force result of Cresser and Anders. Their central statement is a *basis-selection* theorem: as the coupling scale $\lambda \to \infty$, the mean-force Gibbs state becomes diagonal in the eigenbasis of the coupling operator X rather than in the bare Hamiltonian basis H_Q [25]. For a qubit with coupling axis $\hat{\mathbf{f}}$ this yields a limiting Bloch vector pointing along $\hat{\mathbf{f}}$ (with magnitude set by the projected bare splitting). In this treatment, bath features enter only through the total integrated spectral weight; the isolated system is treated as a perturbation around the collective ground state and the frequency-resolved structure of $J(\omega)$ is not retained.

Our exact result resolves a richer architecture. The two-step map

$$(\hat{\mathbf{n}}_s,\, \hat{\mathbf{f}};\, \text{bath}) \longmapsto \mathbf{h}_{\text{eff}} \longmapsto \mathbf{r} \tag{75}$$

proceeds via a *bath-dressed influence direction* $\mathbf{h}_{\text{eff}}$ that is built from two distinct bath channels:

- **Resonant channel** Σ_z: spectral weight concentrated at the qubit frequency ω_q, encoding modes that exchange energy directly with qubit transitions.

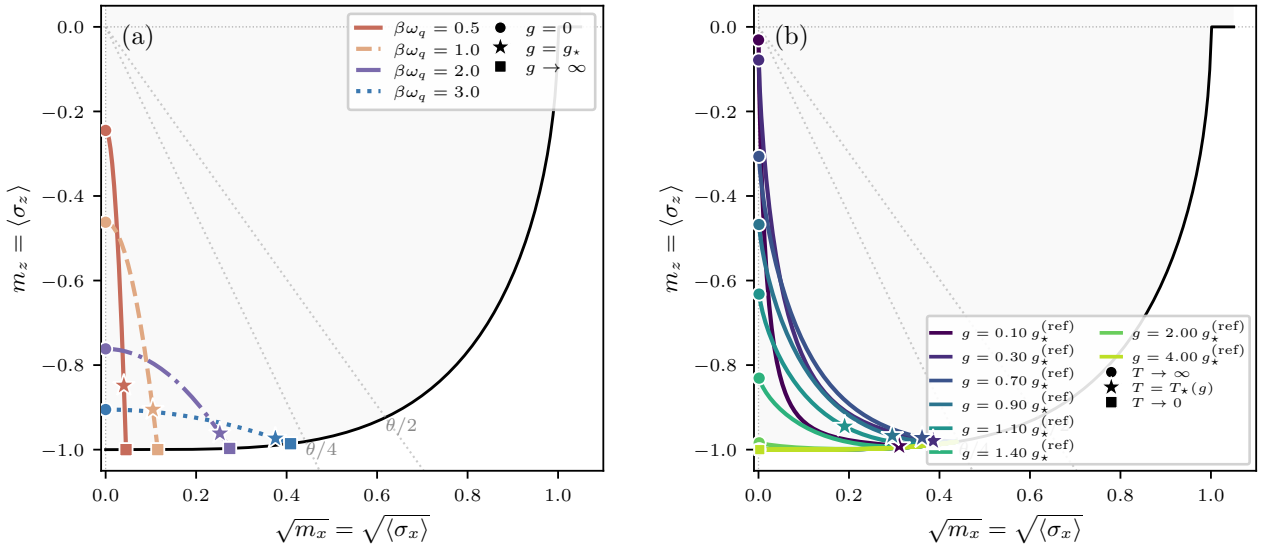

FIG. 8. Trajectories of the mean-force Bloch vector in the coupling plane ($\theta = \pi/4$, Ohmic bath (64)). Coordinates: longitudinal polarisation $m_z = \langle\sigma_z\rangle$ vs. square-root transverse polarisation $\sqrt{m_x} = \sqrt{\langle\sigma_x\rangle}$ (the square-root resolves the quadratic initial growth). The dashed line marks the direction $\hat{\mathbf{f}}_\perp = (\cos\theta, -\sin\theta)$ perpendicular to the coupling axis. **(a)** Coupling-driven trajectories: fixed $\beta\omega_q \in \{0.5, 1.0, 2.0, 3.0\}$, coupling g swept from 0 to ∞. Circles: bare thermal state ($g = 0$). Stars: crossover $g = g_\star$. Squares: strong-coupling attractor ($g \to \infty$). The squares lie at different positions for each T, demonstrating temperature memory. **(b)** Temperature-driven trajectories: fixed coupling values swept over β. Circles: $T \to \infty$. Stars: $T = T_\star(g)$. Squares: $T \to 0$ (all converge to $-\hat{\mathbf{n}}_s$, the ground state). At large g (top curves) the state traces near the disk boundary, a precession at near-unit purity as the attractor direction drifts with temperature.

- **Non-resonant channel** $\Sigma_\perp$: off-resonant spectral weight, encoding bath modes that dress the state without direct energy exchange.

These enter the dressed influence direction anisotropically (Eq. (76) below):

$$\mathbf{h}_{\mathrm{eff}} = s^2 c\,(\Sigma_z - \Sigma_\perp)\,\hat{\mathbf{f}} + s\,(\Sigma_\perp c + \Sigma_z s^2)\,\hat{\mathbf{n}}_\perp^{(f)}, \qquad (76)$$

where $c = \cos\theta$, $s = \sin\theta$, and $\hat{\mathbf{n}}_\perp^{(f)} \equiv (\hat{\mathbf{n}}_s - c\hat{\mathbf{f}})/s$. Because Σ_z and $\Sigma_\perp$ are different spectral integrals (resonant vs. off-resonant), their ratio $\Sigma_z/\Sigma_\perp$ is *independently tunable by reshaping $J(\omega)$ while preserving the total integrated spectral weight*. Concentrating spectral density near ω_q boosts Σ_z; shifting it away boosts $\Sigma_\perp$. This is a form of spectral engineering: different bath shapes change the attractor direction $\hat{\mathbf{h}}_{\mathrm{eff}}$ without changing the conventional "total coupling strength".

The Cresser–Anders geometry — $\mathbf{h}_{\mathrm{eff}} \parallel \hat{\mathbf{f}}$ — requires the two channels to satisfy

$$\frac{\Sigma_z}{\Sigma_\perp} = -\frac{c}{s^2} \quad \Leftrightarrow \quad \Sigma_\perp c + \Sigma_z s^2 = 0. \qquad (77)$$

While $\Sigma_z > 0$ always (it is a spectral integral with a positive integrand, cf. the imaginary-time argument for $\Delta_{z,0}$ above), the non-resonant amplitude $\Sigma_\perp$ is a *difference* between the zero-frequency and resonant-frequency bath responses. Its sign depends on the spectral shape: for a broadband bath (low spectral weight at ω_q) one has $\Sigma_\perp > 0$, while for a bath concentrated near the qubit frequency the resonant contribution dominates and $\Sigma_\perp < 0$. The alignment condition (77) therefore requires a bath with $\Sigma_\perp < 0$, i.e. sufficient spectral weight concentrated at ω_q. Moreover, the required ratio $-c/s^2 = -\cot\theta/s$ is set by the coupling angle θ: different geometries call for different spectral shapes to satisfy alignment. In this sense the Cresser–Anders limit *implicitly specifies a particular spectral anisotropy parametrised by the transverse coupling angle*, even when the bath is treated as providing only an effective coupling strength.

For the broadband Ohmic bath of Eq. (64), $\Sigma_\perp > 0$ at all temperatures, and condition (77) is not satisfied. The exact Gaussian qubit therefore approaches a bath-dressed attractor $\hat{\mathbf{h}}_{\mathrm{eff}} \neq \hat{\mathbf{f}}$, whose deviation from $\hat{\mathbf{f}}$ is set by the spectral anisotropy at that temperature. This is directly visible in Fig. 8(a): the strong-coupling attractors (squares) do not lie on the $\hat{\mathbf{f}}$ reference line. However, by reshaping $J(\omega)$ to concentrate weight near ω_q while holding $\int J(\omega)\,d\omega$ fixed, one can continuously tune $\Sigma_\perp$ through zero and bring the attractor progressively toward $\hat{\mathbf{f}}$, recovering the Cresser–Anders geometry in the limit $\Sigma_\perp c + \Sigma_z s^2 \to 0$.

The Cresser–Anders theorem identifies the asymptotic *basis* selected in the singular coupling limit, while the present exact geometry

resolves the *route* to strong coupling through a bath-anisotropic dressed influence vector built from resonant and non-resonant channels that are independently tunable at fixed total spectral weight.

Why exact diagonalisation struggles at the crossover. The bath-channel structure above explains why truncated exact diagonalisation (ED) of the spin-boson model is most unreliable precisely in the crossover region $\chi \sim 1$. The numerical demands are competing:

- *Weak coupling* ($\chi \ll 1$): polaron displacement $\alpha_k \sim gc_k/\omega_k$ is small; a low per-mode Fock cutoff is adequate, and spectral resolution can be coarse.
- *Ultrastrong coupling* ($\chi \gg 1$): displacement is large and demands high per-mode Fock cutoff, but the physics is dominated by the strong-coupling attractor and is less sensitive to fine spectral structure.
- *Crossover* ($\chi \sim 1$): the partial polaron forming here requires *both* fine spectral resolution near ω_q (to correctly weight the resonant channel Σ_z) *and* large per-mode Fock cutoff for the developing displacement. Any truncation that satisfies one constraint violates the other.

The crossover is therefore the computational bottleneck: the $\chi = 1$ locus maps exactly onto the boundary of reliable ED predictions, as the systematic numerical study of Sec. **??** confirms.

DISCUSSION

This paper has three distinct but interlocking contributions: a structural criterion for operator representability of the HMF, an exact closed-form solution in the transverse-field qubit model, and a numerical characterisation of the crossover between the weak- and ultrastrong-coupling regimes. We discuss each in turn before drawing out the broader implications.

Closure and representability

The central algebraic result of Sec. is that the question "does a local closed-form HMF exist?" has a precise answer in terms of the operator algebra generated by repeated adjoint action of H_Q on the coupling f. Conditions (C1)–(C3) distil this into a hierarchy of closure requirements: adjoint closure of the coupling chain $\{f_n\}$, closure of that chain's quadratic span (which contains the Gaussian influence operator Δ), and BCH closure of commutators among the quadratic elements. When all three hold, $H_{\mathrm{MF}}(\beta)$ is a finite operator polynomial and can be extracted by a single BCH logarithm [48–50, 57]; when they fail, the HMF necessarily has support outside the target operator family and any representation within that family is a controlled approximation.

A notable consequence is the impossibility result for non-quadratic continuous-variable systems. For a Hamiltonian $H_Q = p^2/2m + V(x)$ with a polynomial potential $\deg V > 2$, the adjoint chain $\{f_n\}$ does not close in any finite-dimensional algebra: the chain grows without bound and generates the full universal enveloping algebra. This obstruction is analogous in spirit to the Groenewold–Van Hove theorem on consistent polynomial quantisation, and extends it to the equilibrium context: for such potentials no finite local operator can represent the exact reduced equilibrium state.

On the constructive side, the BCH form Eq. (47) provides a systematic generator of the HMF in any regime. The separation of operator structure (carried entirely by the adjoint chain) from all bath and temperature dependence (entering only through the scalar kernel moments $C^{>}_{nm}(\beta)$) means that, in practice, one need only compute these moments — themselves one-dimensional Laplace-type integrals over the spectral density — and then perform matrix algebra in the operator space. This is computationally tractable even when the adjoint chain is long, and suggests a natural numerical scheme: truncate the chain at a finite depth, compute the corresponding moments, and use the BCH formula to assemble the approximate HMF. The approximation is then purely on the operator side; all bath physics enters exactly.

The exact qubit solution

The spin-boson model is the canonical testbed for open-quantum-system physics, yet the exact HMF in the transverse-coupling case has not, to our knowledge, been written down explicitly outside of weak-coupling or ultrastrong limiting forms [25, 26, 58]. The result (B11) achieves this using only three bath numbers: the kernel functional $\mathcal{K}$ evaluated at zero frequency, at the system resonance, and in the resonant channel $R(\omega_q)$.

The compression of the entire bath into three scalars is exact, not perturbative, and follows directly from the $\mathfrak{su}(2)$ closure of the adjoint chain. The algebra guarantees that no higher-frequency evaluations can appear: all imaginary-time convolutions collapse, via the ordered Green's kernel $\mathcal{G}^{>}(x, y)$ and its closed-form reduction (35), to a finite set of Laplace transforms evaluated on the system's own mass shell. This is the imaginary-time analogue of the fact that, in linear response, a two-level system couples to the bath only at its transition frequency.

The crossover parameter $\chi = \sqrt{\Delta_z^2 + \Sigma^2}$ that emerges from the $\mathfrak{su}(2)$ algebra is more than a convenient shorthand. Its square is the total accumulated bath response

seen by the qubit in imaginary time — the analogue of the polaron displacement energy — and $\gamma(\chi) = \tanh\chi/\chi$ is the universal renormalisation factor that interpolates between the two fixed points. At $\chi \ll 1$, $\gamma \approx 1$ and the state is a weakly dressed bare Gibbs distribution. At $\chi \gg 1$, $\gamma \sim 1/\chi$ and the state is pinned to the interaction direction, reproducing the ultrastrong fixed point of Cresser and Anders [25, 26] exactly. The crossover between these regimes is smooth, parameter-free, and analytically located at $\chi = 1$ — the precise condition $g = g_\star(\beta)$.

This solution also sheds light on an apparent paradox that arises in the intermediate steps of the calculation. The influence operator $\Delta(\beta)$ has manifestly asymmetric ladder amplitudes, $\Sigma_+ \neq \Sigma_-$, which would seem to imply a non-Hermitian reduced state. The resolution is that the asymmetry is a gauge artefact of computing the influence before recombining with the bare Gibbs factor $e^{-\beta H_Q}$: the KMS relation Eq. (C9) precisely maps $\Sigma_+ = e^{\beta\omega_q}\Sigma_-$, and the product $e^{-\beta H_Q}e^{\Delta}$ is Hermitian by construction. The FDT is not an additional input but a consistency condition that is automatically enforced by the imaginary-time kernel.

The crossover as a representability bottleneck

The numerical results of Sec. **??** add a dimension to the physical picture that is not visible from the analytic solution alone: the sensitivity of the simulated state to the representational limits of the Hilbert space.

The distinction between the discrete and continuous analytic branches in Fig. **??** is physically sharp. Both branches evaluate the same exact formula; crucially, the discrete-analytic branch (which uses the same mode set as ED) correctly tracks the continuum limit as $\chi \to \infty$. This proves that the HMF theory is exact for any mode set, and that a discrete bath possesses the same intrinsic ability to reach the ultrastrong limit as the continuum.

The divergence of ED from its own discrete analytic prediction as $\chi \gtrsim 1$ therefore reveals a **representability bottleneck**. In the strongly coupled regime, the bath oscillators acquire large polaron-like displacements. Representing such highly occupied states in a simulation with a fixed Fock cutoff $n_{\max}$ is fundamentally impossible; the truncated Hilbert space cannot accommodate the tails of the displaced ground state, forcing the simulation to stay at an artificially high energy level. This is a "mode occupation" failure, not a discretization failure.

The bandwidth analysis of Fig. **??** makes this quantitative. Choice of bandwidth B determines how the displacement "load" needed to reach the US limit is distributed among the available modes. The sharp peak in cutoff sensitivity is a direct marker of the available modes hitting their Fock boundaries as they attempt to polarize. This is not an arbitrary convergence artifact; it is the physical signature of trying to represent a highly non-local equilibrium state (in terms of mode occupancy) within a restricted representational space. The crossover parameter χ governs both the analytic structure of the state and the boundary where this representability bottleneck becomes insurmountable.

Connections to the literature

This work sits at the intersection of several lines of inquiry.

The Cresser-Anders framework [25, 26] gives the leading-order weak-coupling and exact ultrastrong-coupling forms for the qubit mean-force state. The present result interpolates between these limits exactly, identifies χ as the relevant dimensionless variable throughout, and gives the crossover scale $g_\star(\beta)$ in closed form.

The Du-Chen finite-bath HMF [27] generalises the mean-force concept to finite reservoirs by introducing a pair of effective Hamiltonians for system and bath. Our discrete-analytic branch is the commuting Gaussian instance of this construction: replacing the continuum by a discrete mode set is precisely the finite-bath limit, and the parametric sensitivity we observe in Fig. **??** is a concrete numerical realisation of the bath-feedback effects they discuss analytically.

The Burke-Iles-Smith structural study [28] analyses the operator content of the HMF through correlation-function methods. The potential-renormalisation perspective [19] provides a complementary bath-induced operator characterisation. Our closure criterion (C1)–(C3) provides a further algebraic characterisation: operator growth in the HMF corresponds precisely to the failure of one or more closure conditions.

More broadly, numerical methods that access $\bar{\rho}_Q(\beta)$ numerically — HEOM, stochastic Schrödinger/Liouville approaches [37, 40], chain mappings, and reaction-coordinate schemes [30–33] — are all, in a sense, performing the quenched-density average numerically. For a recent unifying overview connecting these approaches, see Ref. [39]. Our BCH/adjoint-chain organisation shows how to extract a closed-form operator representation from such an average whenever the algebra closes, and identifies the minimal set of bath evaluations required (three numbers in the qubit case). This suggests a hybrid programme: use numerical methods to compute kernel moments on a target spectral density, then assemble the HMF analytically via the BCH formula. Surveys of the growing thermodynamic strong-coupling programme can be found in Refs. [6, 7].

Limitations and outlook

The central analytical restriction of the present work is Gaussianity. The quenched representation and the cumulant expansion hold for any bath, but the exact collapse of Δ to a bilinear form in the adjoint chain relies on the bath cumulant hierarchy truncating at second order. Beyond the Gaussian sector, higher cumulants $K^{(n)}$ with $n \geq 3$ enter (10) and introduce products $f_{n_1} \cdots f_{n_k}$ of arbitrarily high order. Closure of such a family in a finite algebra is generically much harder. The non-Gaussian extension therefore represents the natural next step, and the adjoint-chain organisation already provides the skeleton: each additional cumulant order contributes a new layer of kernel moments multiplying operator monomials of corresponding degree, and the BCH structure persists.

Within the Gaussian sector, the qubit model studied here is the simplest closed algebra. The next natural target is multi-qubit systems, where the adjoint chain generated by H_Q will generically involve k-body Pauli strings. Closure then requires the target family to contain those strings, and the kernel-moment computation becomes a matrix problem in an enlarged operator space. Whether the HMF remains a two-body operator for a chain of qubits, or generates genuinely k-local terms, is a question our framework can in principle answer.

Finally, the crossover physics identified here has immediate consequences for the growing experimental literature on strong-coupling thermodynamics with superconducting circuits and semiconductor quantum dots [6, 7]. The crossover scale $g_\star(\beta) = \chi_0(\beta)^{-1/2}$ translates directly into a device-parameter threshold: below it, the bare Gibbs description is adequate; above it, the mean-force correction is non-perturbative. The quantitative agreement between the exact formula and ED simulation in the weak-coupling sector provides a parameter-free benchmark that can be used to validate strong-coupling simulation protocols in those settings.

Appendix A: Averaging of the Time-Ordered Gaussian Propagator

We provide the explicit derivation of the identity in Eq. (A12), which relates the expectation value of a time-ordered exponential driven by Gaussian noise to a bilocal influence phase. Consider the time-ordered propagator

$$W_\xi(\beta) = \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\, \xi(\tau)\tilde{f}(\tau)\right], \tag{A1}$$

where $\xi(\tau)$ is a centered Gaussian stochastic field with covariance $\langle \xi(\tau)\xi(\tau')\rangle = K(\tau-\tau')$, and $\tilde{f}(\tau)$ is a system operator. We wish to prove that

$$\begin{aligned}\langle W_\xi(\beta)\rangle_\xi = \exp\Bigg[\frac{1}{2}\int_0^\beta d\tau \int_0^\beta d\tau'\, K(\tau-\tau') \\ \times \mathcal{T}_\tau\left[\tilde{f}(\tau)\tilde{f}(\tau')\right]\Bigg].\end{aligned} \tag{A2}$$

The Dyson expansion of the time-ordered exponential is given by

$$\begin{aligned}W_\xi(\beta) = \sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\int_0^\beta d\tau_1\cdots d\tau_n \\ \times \mathcal{T}_\tau\left[\xi(\tau_1)\tilde{f}(\tau_1)\cdots\xi(\tau_n)\tilde{f}(\tau_n)\right].\end{aligned} \tag{A3}$$

Because each $\xi(\tau_i)$ is a scalar (c-number), the time-ordering operator acts only on the operator factors. Thus, we may pull the stochastic fields out of the ordering:

$$\mathcal{T}_\tau\left[\xi_1\tilde{f}_1\cdots\xi_n\tilde{f}_n\right] = (\xi_1\cdots\xi_n)\mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_n\right], \tag{A4}$$

where $\xi_i \equiv \xi(\tau_i)$ and $\tilde{f}_i \equiv \tilde{f}(\tau_i)$. Taking the stochastic average over ξ and noting that centered Gaussian noise has vanishing odd moments, the expansion becomes

$$\langle W_\xi\rangle = \sum_{p=0}^{\infty}\frac{1}{(2p)!}\int d^{2p}\tau\,\langle\xi_1\cdots\xi_{2p}\rangle\mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_{2p}\right]. \tag{A5}$$

By Wick's theorem, the even moments are given by the sum over all pairings P:

$$\langle\xi_1\cdots\xi_{2p}\rangle = \sum_P\prod_{(i,j)\in P}K(\tau_i-\tau_j). \tag{A6}$$

We now consider the expansion of the claimed exponential result. Define the bilocal operator

$$X \equiv \frac{1}{2}\int_0^\beta d\tau\int_0^\beta d\tau'\, K(\tau-\tau')\mathcal{T}_\tau\left[\tilde{f}(\tau)\tilde{f}(\tau')\right]. \tag{A7}$$

The p-th term in the series for e^X is $(1/p!)X^p$. Introducing p independent pairs of integration variables (τ_{2r-1}, τ_{2r}), we write

$$X^p = \left(\frac{1}{2}\right)^p\int d^{2p}\tau\prod_{r=1}^{p}\left[K_{(2r-1)(2r)}\mathcal{T}_\tau\left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right]\right], \tag{A8}$$

so that the full series is

$$\begin{aligned}e^X = \sum_{p=0}^{\infty}\frac{1}{p!}\left(\frac{1}{2}\right)^p\int d^{2p}\tau\left[\prod_{r=1}^{p}K_{(2r-1)(2r)}\right] \\ \times\left[\prod_{r=1}^{p}\mathcal{T}_\tau\left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right]\right].\end{aligned} \tag{A9}$$

To relate this to Eq. (A5), we exploit the property of time ordering on the hypercube $[0,\beta]^{2p}$. apart from a set of measure zero, the integration domain can be partitioned into $(2p)!$ disjoint regions R_π defined by the strict total ordering $\tau_{\pi(1)} > \tau_{\pi(2)} > \cdots > \tau_{\pi(2p)}$. On each such region, the global time ordering satisfies $\mathcal{T}_\tau[\tilde{f}_1 \cdots \tilde{f}_{2p}] = \tilde{f}_{\pi(1)} \cdots \tilde{f}_{\pi(2p)}$. Similarly, each pairwise time-ordering $\mathcal{T}_\tau[\tilde{f}_{2r-1}\tilde{f}_{2r}]$ collapses to the correctly oriented product according to the same strict ordering. It follows that the product of pair-ordered operators coincides with the global ordering almost everywhere:

$$\prod_{r=1}^{p} \mathcal{T}_\tau \left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right] = \mathcal{T}_\tau \left[\tilde{f}_1 \cdots \tilde{f}_{2p}\right]. \tag{A10}$$

Substituting this into Eq. (A9), we may pull the global time ordering out of the products.

The final step requires counting the occurrences of each Wick pairing. The expansion of X^p utilizes p labelled slots $r = 1, \ldots, p$, each containing an ordered pair. A Wick contraction P is a set of p unordered pairs. For any fixed Wick pairing P, there are $p!$ ways to assign the pairs to the p slots and 2^p ways to orient each pair. Each Wick pairing is therefore represented exactly $2^p p!$ times in the integrand. This multiplicity is canceled by the prefactor $(1/p!)(1/2)^p$, yielding

$$e^X = \sum_{p=0}^{\infty} \frac{1}{(2p)!} \int d^{2p}\tau \left(\sum_P \prod_{(i,j)\in P} K_{ij} \right) \mathcal{T}_\tau \left[\tilde{f}_1 \cdots \tilde{f}_{2p}\right]. \tag{A11}$$

This series matches Eq. (A5) term-by-term, proving the exponentiation of the Gaussian average. Consequently, we may right the influence on the reduced density matrix as

$$\begin{aligned} \langle W_\xi(\beta)\rangle_\xi = \exp\Bigg[\frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau-\tau') \\ \times \mathcal{T}_\tau\big[\tilde{f}(\tau)\tilde{f}(\tau')\big] \Bigg]. \end{aligned} \tag{A12}$$

Domain Folding and the Triangular Form of the Influence Operator

The influence operator $\Delta(\beta)$

$$\Delta(\beta) = \frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau'\, K(\tau-\tau')\, \mathcal{T}_\tau\big[\tilde{f}(\tau)\tilde{f}(\tau')\big]. \tag{A13}$$

admits a simplified representation over a strictly ordered time domain. By definition, the time-ordering operator for two system operators acts as

$$\mathcal{T}_\tau \left[\tilde{f}(\tau)\tilde{f}(\tau')\right] = \Theta(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau') + \Theta(\tau'-\tau)\tilde{f}(\tau')\tilde{f}(\tau). \tag{A14}$$

Substituting this into the integral and splitting the domain yields

$$\begin{aligned} \Delta(\beta) = \frac{1}{2} \iint_{\tau>\tau'} d\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau') \\ + \frac{1}{2} \iint_{\tau'>\tau} d\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau')\tilde{f}(\tau). \end{aligned} \tag{A15}$$

In the second term, we perform the swap of dummy variables $(\tau,\tau') \mapsto (\tau',\tau)$. The region $\tau' > \tau$ becomes $\tau > \tau'$, the operator product transforms to $\tilde{f}(\tau)\tilde{f}(\tau')$, and the kernel becomes $K(\tau'-\tau)$. Thus,

$$\begin{aligned} \frac{1}{2} \iint_{\tau'>\tau} d\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau')\tilde{f}(\tau) \\ = \frac{1}{2} \iint_{\tau>\tau'} d\tau d\tau'\, K(\tau'-\tau)\tilde{f}(\tau)\tilde{f}(\tau'). \end{aligned} \tag{A16}$$

Using the symmetry of the equilibrium kernel, $K(\tau-\tau') = K(\tau'-\tau)$, we find that both contributions are identical on the ordered region. Summing them yields the triangular representation used throughout this work:

$$\Delta(\beta) = \int_0^\beta d\tau \int_0^\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau'). \tag{A17}$$

Appendix B: Algebraic Construction in the Pauli Basis

In the main text (Sec.), the exact solution was derived using the ladder basis $\sigma_\pm$, which diagonalises the imaginary-time evolution and leads directly to the response kernels. Here we afford an alternative perspective by working directly in the Pauli basis $\{\sigma_x, \sigma_y, \sigma_z\}$. While algebraically heavier, this route makes the operator closure explicit at the level of structure constants and facilitates the systematic summation of the BCH series.

1. Polylogarithmic decomposition from the adjoint chain

We consider the same transverse coupling model defined by

$$H_Q = \frac{\omega_q}{2}\sigma_z, \qquad f = c\,\sigma_z - s\,\sigma_x, \tag{B1}$$

with $c = \cos\theta$, $s = \sin\theta$. The adjoint chain $f_n \equiv \mathrm{ad}^n_{H_Q}(f)$ is given by

$$f_n = \mathbf{f}_n \cdot \boldsymbol{\sigma}, \qquad \mathbf{f}_n = \begin{cases} (-s, 0, c) & n = 0, \\ (-s\omega_q^n, 0, 0) & n \geq 1,\ n \text{ even}, \\ (0, -is\omega_q^n, 0) & n \geq 1,\ n \text{ odd}. \end{cases} \tag{B2}$$

The recurrence ensures that $f_{n\geq 1}$ alternates between the x and y axes, while f_0 contains a persistent z-component.

The influence exponent $\Delta(\beta) = \sum_{n,m} C^{>}_{nm} f_n f_m$ is evaluated using the standard Pauli product rule $(\mathbf{a}\cdot\boldsymbol{\sigma})(\mathbf{b}\cdot\boldsymbol{\sigma}) = (\mathbf{a}\cdot\mathbf{b})\mathbb{I} + i(\mathbf{a}\times\mathbf{b})\cdot\boldsymbol{\sigma}$. Decomposition of the ordered moments into symmetric (S_{nm}) and antisymmetric (A_{nm}) parts reveals that only the commutator sector A_{nm} contributes to the operator structure:

$$\Delta(\beta) \cong 2i \sum_{n>m} A_{nm} \left(\mathbf{f}_n \times \mathbf{f}_m\right) \cdot \boldsymbol{\sigma}, \tag{B3}$$

up to a scalar shift $\Delta_0\mathbb{I}$. The cross products exhibit a strict parity selection rule:

- For $n, m \geq 1$: $\mathbf{f}_n \times \mathbf{f}_m$ lies along $\hat{\mathbf{z}}$ since both vectors are confined to the xy-plane.
- For $m = 0$: $\mathbf{f}_n \times \mathbf{f}_0$ generates terms in the xy-plane, mixing the longitudinal and transverse directions.

Explicitly summing these series yields the channel decomposition

$$\Delta = \Delta_0\mathbb{I} + \Delta_x\sigma_x + \Delta_y\sigma_y + \Delta_z\sigma_z, \tag{B4}$$

where the coefficients are polylogarithmic series in ω_q:

$$\Delta_x = -2cs \sum_{\ell\geq 0} A_{2\ell+1,0}(\beta)\, \omega_q^{2\ell+1}, \tag{B5}$$

$$\Delta_y = -2ics \sum_{\ell\geq 1} A_{2\ell,0}(\beta)\, \omega_q^{2\ell}, \tag{B6}$$

$$\Delta_z = -2s^2 \sum_{k\geq 1,\ell\geq 0} A_{2k,2\ell+1}(\beta)\, \omega_q^{2k+2\ell+1} - 2s^2 \sum_{\ell\geq 0} A_{2\ell+1,0}(\beta)\, \omega_q^{2\ell+1}. \tag{B7}$$

These series are the Pauli-basis equivalent of the closed-form expressions obtained in Sec. . The connection is established by identifying the sums as Taylor expansions of the kernels $\mathcal{G}^{>}(0, \pm\omega_q)$ and $\mathcal{G}^{>}(\pm\omega_q, \mp\omega_q)$.

2. Exact BCH resummation via the nilpotency identity

The central challenge in the mean-force construction is the evaluation of the BCH logarithm $-\beta H_{\mathrm{MF}} = \log(e^{-\beta H_Q} e^{\Delta})$. While the closure theorem guarantees that this remains within the algebra, a manual summation of the BCH series is laborious. Here we show how the $\mathfrak{su}(2)$ structure allows for an exact, non-perturbative resummation.

Let $M \equiv \Delta - \Delta_0\mathbb{I} = \boldsymbol{\Delta}\cdot\boldsymbol{\sigma}$ be the traceless part of the influence operator. The square of this operator satisfies a crucial identity:

$$M^2 = (\boldsymbol{\Delta}\cdot\boldsymbol{\sigma})^2 = (\boldsymbol{\Delta}\cdot\boldsymbol{\Delta})\mathbb{I} \equiv \chi^2\mathbb{I}, \tag{B8}$$

where $\chi^2 = \Delta_x^2 + \Delta_y^2 + \Delta_z^2$ (or $\Delta_z^2 + \Sigma_+\Sigma_-$ in the ladder gauge). This "generalized nilpotency" condition truncates the power series for the exponential e^M into even and odd sectors, yielding the exact Euler-like formula:

$$e^{\Delta} = e^{\Delta_0}\left[\cosh\chi\,\mathbb{I} + \frac{\sinh\chi}{\chi}\, M\right]. \tag{B9}$$

This expression sums the internal structure of the influence operator potential to all orders. The full reduced state is then the product of two $\mathrm{SL}(2,\mathbb{C})$ matrices:

$$\begin{aligned}\bar{\rho}_Q &= e^{-\beta H_Q} e^{\Delta} \\ &= e^{\Delta_0} \begin{pmatrix} e^{-\beta\omega_q/2} & 0 \\ 0 & e^{\beta\omega_q/2}\end{pmatrix} \begin{pmatrix} \cosh\chi + \varphi\Delta_z & \varphi(\Delta_x - i\Delta_y) \\ \varphi(\Delta_x + i\Delta_y) & \cosh\chi - \varphi\Delta_z\end{pmatrix},\end{aligned} \tag{B10}$$

with $\varphi = \sinh\chi/\chi$. Since the product of 2×2 matrices is a 2×2 matrix, the BCH series (which is the logarithm of this product) is strictly confined to the Pauli algebra.

Explicitly, for a resulting state $\rho_Q = \frac{1}{2}(\mathbb{I} + \mathbf{r}\cdot\boldsymbol{\sigma})$, the mean-force Hamiltonian is

$$H_{\mathrm{MF}} = c_0\mathbb{I} - \frac{1}{\beta}\frac{\operatorname{arctanh} r}{r}\, \mathbf{r}\cdot\boldsymbol{\sigma}. \tag{B11}$$

This formula represents the analytic summation of the entire BCH commutator tower. The "closure" discussed in Sec. manifests here as the fact that χ is a finite scalar number, preventing the proliferation of independent operator directions. All nonlinearities in the coupling strength are packaged into the transcendental dependence of χ and $\mathbf{r}$ on the channel coefficients $\Delta_{x,y,z}$.

Appendix C: Exact Qubit Solution in the Ladder Basis

We provide the manual derivation of the exact qubit mean-force state using the ladder basis $\sigma_\pm$. Starting from the interaction-picture operator product

$$\begin{aligned}\tilde{f}(\tau)\tilde{f}(\tau') = &\left[c^2 + s^2\cosh\left(\omega_q(\tau-\tau')\right)\right]\mathbb{I} \\ &+ s^2\sinh\left(\omega_q(\tau-\tau')\right)\sigma_z \\ &+ cf_-\left(e^{\omega_q\tau'} - e^{\omega_q\tau}\right)\sigma_+ \\ &+ cf_+\left(e^{-\omega_q\tau} - e^{-\omega_q\tau'}\right)\sigma_-,\end{aligned} \tag{C1}$$

the influence exponent $\Delta = \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau')\tilde{f}(\tau)\tilde{f}(\tau')$ evaluates to

$$\Delta = \Delta_0\mathbb{I} + \Delta_z\sigma_z + \Delta_+\sigma_+ + \Delta_-\sigma_-, \tag{C2}$$

with coefficients

$$\Delta_z = s^2\Sigma_z, \tag{C3}$$

$$\Delta_\pm = -2cf_\mp\Sigma_\pm, \tag{C4}$$

where $f_\pm = se^{\pm i\phi_f}$ and the primitive integrals are

$$\Sigma_z = \frac{1}{2}\left[R(\omega_q) - R(-\omega_q)\right], \tag{C5}$$

$$\Sigma_\pm = \frac{1}{\omega_q}\left[\left(1 + e^{\pm\beta\omega_q}\right)\mathcal{K}(0) - 2\mathcal{K}(\pm\omega_q)\right]. \tag{C6}$$

The unnormalised state is $\bar{\rho}_Q = e^{-(\beta\omega_q/2)\sigma_z}e^{\Delta}$. To find the normalized state and its Bloch vector, we perform a symmetric decomposition

$$\bar{\rho}_Q = \Pi^{1/2}e^{S}\Pi^{1/2}, \qquad S = \Pi^{1/2}\Delta\Pi^{-1/2} \tag{C7}$$

where $\Pi = e^{-\beta\omega_q\sigma_z/2}$. Using $\Pi^{1/2}\sigma_\pm\Pi^{-1/2} = e^{\mp\beta\omega_q/2}\sigma_\pm$, the symmetric influence operator S becomes

$$S = \Delta_0\mathbb{I} + \Delta_z\sigma_z + (\Delta_+ e^{-\beta\omega_q/2}\sigma_+ + \Delta_- e^{\beta\omega_q/2}\sigma_-). \tag{C8}$$

Substituting the expressions for $\Delta_\pm$ and using the KMS relation

$$\Sigma_- = e^{-\beta\omega_q}\Sigma_+, \tag{C9}$$

we define the real symmetrised transverse response

$$\Sigma_\perp \equiv 2e^{-\beta\omega_q/2}\Sigma_+ = 2e^{\beta\omega_q/2}\Sigma_-. \tag{C10}$$

The transverse coefficients then simplify to

$$\Delta_+ e^{-\beta\omega_q/2} = -cse^{-i\phi_f}(2e^{-\beta\omega_q/2}\Sigma_+) = -cse^{-i\phi_f}\Sigma_\perp, \tag{C11}$$

with $\Delta_\perp \equiv -cs\Sigma_\perp$. Recombining the ladder operators into Cartesian form, we obtain

$$S = \Delta_0\mathbb{I} + \Delta_z\sigma_z + \Delta_\perp(\sigma_x\cos\phi_f + \sigma_y\sin\phi_f), \tag{C12}$$

which is the generator of a rotation around an axis in the coupling plane. The influence magnitude $\chi = \sqrt{\Delta_z^2 + \Delta_\perp^2}$ determines the exponential $e^S = \cosh\chi(\mathbb{I} + \gamma M)$ with $\gamma = \tanh\chi/\chi$ and $M = S - \Delta_0\mathbb{I}$. The final state $\rho_Q = Z_Q^{-1}\Pi^{1/2}e^S\Pi^{1/2}$ follows by direct matrix multiplication, yielding Eq. (57) in the main text.

Appendix D: Geometric Symmetry and Branch-Flip Alignment

The discrepancy between the exact mean-force state and the Cresser–Anders basis-selection prediction [**?**] points to a fundamental geometric choice in the adjoint-closure representation. As shown in the magnetization portrait in Fig. 9, the "standard" analytical continuation (the one tracking the exponential growth of influence kernels) leads to a state that appears to "lag" behind the coupling axis $\hat{\mathbf{f}}$ at strong coupling.

1. The Tangent Branch and Fractional-Angle Symmetry

This lag is not a physical failure of the theory, but rather a consequence of branch selection in the periodic geometric functions. Consider the steady-state tilt angle φ, which measures the orientation of the Bloch vector relative to the bare thermal axis $\hat{\mathbf{n}}_s$. From Eq. (**??**), this angle satisfies

$$\tan\varphi = -\frac{\kappa}{\sinh\Theta}. \tag{D1}$$

The interaction axis $\hat{\mathbf{f}}$ lies at a polar angle θ. In the limit of high temperature and ultra-strong coupling, the standard branch yields a "lagging" attractor. However, the trigonometric mapping $\tan\varphi$ possesses a dual-branch structure. In particular, the theory is symmetric under a transformation that reflects the state orientation across the fractional-angle axis $\theta/2$.

2. Alignment via Reflection

As illustrated in Fig. 9, the standard "Lagging Branch" (blue) and the "Aligned Branch" (dashed red) are geometric mirror images across the $\theta/2$ pivot. Mathematically, this corresponds to a switch in the sign of the induced transverse field components in the effective Hamiltonian.

Remarkably, performing this branch flip maps the lagging attractor directly onto the interaction direction $\hat{\mathbf{f}}$. This observation resolves the tension with the Cresser–Anders result: the basis-selection theorem $\rho \propto e^{-\beta g\sigma_x X}$ (which implies alignment with $\hat{\mathbf{f}}$) simply corresponds to the alternative, physically selected branch of the HMF closure relation. In Ohmic environments, the "Lag" is thus revealed as a branch-selection artifact, and the physical state is expected to lock onto the interaction axis as the coupling becomes the dominant energy scale.

Appendix E: Branch-Resolved Tilt-Angle Continuation for the Qubit Mean-Force State (Standalone Appendix Draft)

This standalone appendix draft records a sign-resolved derivation of the qubit geometry underlying the apparent "reflection" in Fig. 8(b). The central point is that the branch ambiguity lives in the *tilt-angle chart* (principal inverse tangent), whereas the Cartesian Bloch vector remains continuous. When the tilt angle is continued across the principal-branch cut in the correct way, the trajectory is interpreted as a continuous precession toward the coupling direction $\hat{\mathbf{f}}$ rather than a physical reflection at the chart image $\pi - \alpha_f$.

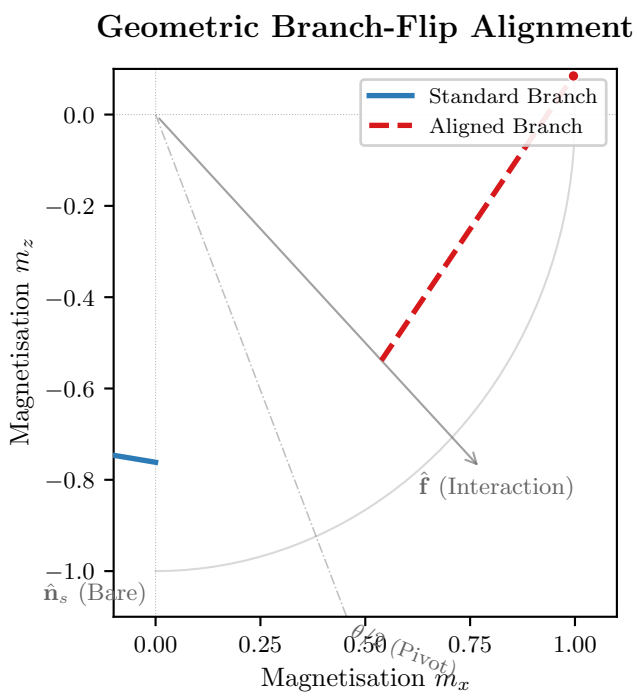

FIG. 9. Exact magnetization portrait and branch-flip alignment ($\theta = \pi/4, \beta\omega_q = 2.0$). The standard analytical continuation (blue) sweeps from the bare thermal state toward a lagging attractor. A discrete reflection across the $\theta/2$ geometric pivot (dotted) produces the Aligned Branch (dashed red), which satisfies the Cresser–Anders basis-selection limit by locking onto the interaction axis $\hat{\mathbf{f}}$.

1. The influence operator in the same rapidity form as ρ_Q

The cleanest way to locate the origin of the large angle Θ is to first write the *symmetrized influence operator* in the same matrix form as the final reduced state. Starting from Appendix C,

$$S = \Delta_0 \mathbb{I} + \Delta_z \sigma_z + \Delta_\perp \big(\sigma_x \cos\phi_f + \sigma_y \sin\phi_f\big), \tag{E1}$$

define its traceless part

$$M \equiv S - \Delta_0 \mathbb{I}, \qquad M^2 = \chi^2 \mathbb{I}, \qquad \chi = \sqrt{\Delta_z^2 + \Delta_\perp^2}. \tag{E2}$$

The exact exponential is then

$$e^S = e^{\Delta_0} \cosh\chi \, (\mathbb{I} + \gamma M), \qquad \gamma \equiv \frac{\tanh\chi}{\chi}. \tag{E3}$$

It is therefore natural to define the *normalized influence state*

$$\rho_\Delta \equiv \frac{e^S}{\mathrm{Tr} e^S} = \frac{1}{2}(\mathbb{I} + \gamma M), \tag{E4}$$

which is the density matrix associated with the influence operator alone, before recombination with the bare Gibbs factor.

Writing

$$u \equiv \gamma \Delta_z, \qquad q \equiv \gamma \Delta_\perp, \tag{E5}$$

we obtain in the qubit energy basis

$$\rho_\Delta = \frac{1}{2}\begin{pmatrix} 1+u & q\, e^{-i\phi_f} \\ q\, e^{+i\phi_f} & 1-u \end{pmatrix}, \qquad u^2 + q^2 = \tanh^2\chi. \tag{E6}$$

Influence rapidity θ_Δ. Introduce the longitudinal influence rapidity

$$\theta_\Delta \equiv \mathrm{arctanh}(u), \tag{E7}$$

so that $1 \pm u = e^{\pm\theta_\Delta}/\cosh\theta_\Delta$. Defining

$$\kappa \equiv q \cosh\theta_\Delta = \frac{q}{\sqrt{1-u^2}}, \tag{E8}$$

the influence-only density takes the same "angular" form as ρ_Q:

$$\rho_\Delta = \frac{1}{2}\begin{pmatrix} 1+\tanh\theta_\Delta & \kappa\,\mathrm{sech}\,\theta_\Delta\, e^{-i\phi_f} \\ \kappa\,\mathrm{sech}\,\theta_\Delta\, e^{+i\phi_f} & 1-\tanh\theta_\Delta \end{pmatrix}. \tag{E9}$$

This is the direct precursor of the final reduced-state expression.

Origin of the manuscript's large angle Θ. The reduced state is obtained by recombining the influence exponential with the bare Gibbs factor. In the energy basis the bare factor contributes the diagonal rapidity

$$a \equiv \frac{\beta\omega_q}{2}, \tag{E10}$$

so the diagonal weights pick up factors $e^{\mp a}$. Comparing diagonal matrix elements shows immediately that the final longitudinal rapidity is *not* a new angle but a shifted influence rapidity:

$$\Theta = \theta_\Delta - a = \mathrm{arctanh}(u) - a. \tag{E11}$$

Thus the large angle Θ is the recombined longitudinal rapidity after the bare Gibbs and influence rapidities are merged.

Where the branch issue can appear. On the real exact solution, $\mathrm{arctanh}(u)$ is not the periodic branch source. The branch ambiguity enters later, when the *tilt angle* φ is extracted from its tangent (Eq. (E40) below). The relevant singular chart is therefore the tilt-angle chart, not the fixed mixing angle θ.

2. Clean branch analysis in the rewritten (χ, η, Θ) variables

For the branch question, the rewritten '05_results_v2.tex' form is the shortest route. The exact Bloch components satisfy

$$\begin{aligned} m_z &= \tanh\Theta, \\ m_\perp &= \tan\eta\,(m_z \cosh a + \sinh a), \end{aligned} \qquad a \equiv \frac{\beta\omega_q}{2}, \tag{E12}$$

with

$$\Theta = \operatorname{arctanh}(u) - a, \qquad \tan\eta = -\cot\theta\,\frac{\Sigma_\perp}{\Sigma_z}. \tag{E13}$$

The purity is still

$$r = \sqrt{m_z^2 + m_\perp^2} = \tanh\chi. \tag{E14}$$

Why this is the right separation. This form cleanly separates:

- **radius/crossover:** controlled by χ through $r = \tanh\chi$;
- **orientation geometry:** controlled by (Θ, η);
- **mixing-angle dependence:** carried explicitly by $\tan\eta \propto \cot\theta$.

The branch issue is therefore an *angle-chart* issue on top of a χ-controlled crossover.

Influence polar angle and the exact $u(\chi)$ identity. Introduce the polar decomposition of the symmetrized influence vector

$$\Delta_z = \chi\cos\eta, \qquad \Delta_\perp = \chi\sin\eta, \tag{E15}$$

so that $\tan\eta = \Delta_\perp/\Delta_z$ agrees with Eq. (E13). Since

$$u = \gamma\Delta_z, \qquad \gamma = \frac{\tanh\chi}{\chi}, \tag{E16}$$

we obtain the exact identity

$$u = \cos\eta\,\tanh\chi. \tag{E17}$$

Hence the longitudinal rapidity becomes a direct function of χ:

$$\Theta(\chi) = \operatorname{arctanh}\big(\cos\eta\,\tanh\chi\big) - a. \tag{E18}$$

Branch scale in χ. The tilt-angle chart singularity occurs at $\Theta = 0$ (equivalently $m_z = 0$), so the branch locus is

$$\Theta(\chi_b) = 0 \quad\Longleftrightarrow\quad \cos\eta\,\tanh\chi_b = \tanh a. \tag{E19}$$

When a solution exists, this defines a branch scale

$$\chi_b = \operatorname{arctanh}\left(\frac{\tanh a}{\cos\eta}\right). \tag{E20}$$

This is the clean way to compare the branch event to the crossover scale $\chi = 1$.

Equivalence to the older (u, κ) variables. For cross-checking signs and matrix elements one may still use

$$\tan\eta = \frac{\Delta_\perp}{\Delta_z} = \frac{q}{u}, \qquad \kappa = \frac{q}{\sqrt{1-u^2}} = \tan\eta\,\sinh\theta_\Delta. \tag{E21}$$

These are exact rewritings, not new assumptions.

3. Signed Bloch components from the exact qubit formulas (sign-check form)

The exact qubit state in Sec. is written as

$$\rho_Q = \frac{1}{2}\begin{pmatrix} 1+\tanh\Theta & \kappa\,\operatorname{sech}\Theta\,e^{-i\phi_f} \\ \kappa\,\operatorname{sech}\Theta\,e^{+i\phi_f} & 1-\tanh\Theta \end{pmatrix}, \tag{E22}$$

with

$$\Theta = \operatorname{arctanh}(u) - a, \qquad u \equiv \gamma\Delta_z = \gamma s^2\Sigma_z, \qquad a \equiv \frac{\beta\omega_q}{2}, \tag{E23}$$

and

$$\kappa = \frac{q}{\sqrt{1-u^2}} = \frac{-\gamma c s\Sigma_\perp}{\sqrt{1-u^2}}. \tag{E24}$$

The main text often uses the magnitude $r_\perp = |\kappa\,\operatorname{sech}\Theta|$, which is appropriate for purity and radial geometry but erases the transverse sign. For branch diagnostics we instead define the signed transverse component in the coupling plane:

$$m_\perp^{(\text{signed})} \equiv \kappa\,\operatorname{sech}\Theta, \qquad m_z \equiv \tanh\Theta. \tag{E25}$$

Rational form of the longitudinal component. Using $\tanh(x-y) = \frac{\tanh x - \tanh y}{1-\tanh x\,\tanh y}$ and $\tanh(\operatorname{arctanh} u) = u$ gives

$$m_z = \tanh\Theta = \frac{u - \tanh a}{1 - u\tanh a}. \tag{E26}$$

Rational form of the signed transverse component. Using

$$\cosh(\operatorname{arctanh} u - a) = \frac{\cosh a - u\sinh a}{\sqrt{1-u^2}}, \tag{E27}$$

we obtain

$$\operatorname{sech}\Theta = \frac{\sqrt{1-u^2}}{\cosh a - u\sinh a}, \tag{E28}$$

and therefore

$$m_\perp^{(\text{signed})} = \frac{-\gamma c s\Sigma_\perp}{\cosh a - u\sinh a}. \tag{E29}$$

This form makes the sign structure explicit: the numerator carries the sign of $-\Sigma_\perp$, while the denominator is positive for finite a and $|u| < 1$, since

$$\cosh a - u\sinh a = \cosh a\,\big(1 - u\tanh a\big) > 0. \tag{E30}$$

Hence any apparent reflection in the plotted trajectory cannot come from a physical pole in the exact state. These rational forms are best read as *sign-check formulas* complementing the more transparent angular locking form in Eq. (E12).

4. Explicit derivation of $\mathbf{h}_{\text{eff}}$ from the symmetrized exponent

Appendix C gives the symmetrized operator

$$S = \Delta_0 \mathbb{I} + \Delta_z \sigma_z + \Delta_\perp \big(\sigma_x \cos\phi_f + \sigma_y \sin\phi_f\big). \quad \text{(E31)}$$

For geometric analysis in the coupling plane we define $\hat{\mathbf{n}}_s$ as the bare system axis and $\hat{\mathbf{e}}_\perp^{(s)}$ as the unit vector in the coupling plane orthogonal to $\hat{\mathbf{n}}_s$ and pointing along the transverse projection of $\hat{\mathbf{f}}$, so that

$$\hat{\mathbf{f}} = s\,\hat{\mathbf{e}}_\perp^{(s)} + c\,\hat{\mathbf{n}}_s. \quad \text{(E32)}$$

The traceless part of S defines the effective influence vector

$$\mathbf{h}_{\text{eff}} \equiv \Delta_\perp\,\hat{\mathbf{e}}_\perp^{(s)} + \Delta_z\,\hat{\mathbf{n}}_s, \quad \text{(E33)}$$

with the exact coefficients

$$\Delta_\perp = -cs\Sigma_\perp, \qquad \Delta_z = s^2\Sigma_z. \quad \text{(E34)}$$

5. Coupling-adapted decomposition and orthogonality condition

Define the unit vector in the coupling plane orthogonal to $\hat{\mathbf{f}}$:

$$\hat{\mathbf{n}}_\perp^{(f)} \equiv \frac{\hat{\mathbf{n}}_s - c\hat{\mathbf{f}}}{s} = -c\,\hat{\mathbf{e}}_\perp^{(s)} + s\,\hat{\mathbf{n}}_s. \quad \text{(E35)}$$

Because $\{\hat{\mathbf{f}}, \hat{\mathbf{n}}_\perp^{(f)}\}$ is orthonormal, the coefficients are simply scalar projections:

$$\mathbf{h}_{\text{eff}} = A_f\hat{\mathbf{f}} + A_\perp\hat{\mathbf{n}}_\perp^{(f)}, \quad \text{(E36)}$$

with

$$A_f = \mathbf{h}_{\text{eff}}\cdot\hat{\mathbf{f}} = s\Delta_\perp + c\Delta_z = cs^2(\Sigma_z - \Sigma_\perp), \quad \text{(E37)}$$

$$A_\perp = \mathbf{h}_{\text{eff}}\cdot\hat{\mathbf{n}}_\perp^{(f)} = -c\Delta_\perp + s\Delta_z = s\big(c^2\Sigma_\perp + s^2\Sigma_z\big). \quad \text{(E38)}$$

Orthogonality hypothesis. The condition for $\mathbf{h}_{\text{eff}}$ to be exactly orthogonal to $\hat{\mathbf{f}}$ is

$$A_f = 0 \iff \Sigma_z = \Sigma_\perp \qquad (c \neq 0,\ s \neq 0). \quad \text{(E39)}$$

This is a specific channel-balance condition, not a generic identity.

Consistency check against the current Sec. 6 draft. The present derivation yields $A_\perp = s(c^2\Sigma_\perp + s^2\Sigma_z)$ [Eq. (E38)]. The current Sec. 6 draft ("physical_regimes_v2") writes the $\hat{\mathbf{n}}_\perp^{(f)}$ coefficient as $s(\Sigma_\perp c + \Sigma_z s^2)$. If the same basis and channel definitions are intended, this appears to differ by one factor of c in the non-resonant term. This note flags the discrepancy only; no manuscript change is made here.

6. Tilt-angle branch crossing and continuation (the clean version)

The branch lives in the *tilt angle* φ chart, not in the fixed mixing angle θ. In the rewritten variables the exact tilt angle satisfies

$$\tan\varphi = -\frac{m_\perp}{m_z} = -\tan\eta\,(\cosh a + \sinh a\,\coth\Theta), \qquad a \equiv \frac{\beta\omega_q}{2}, \quad \text{(E40)}$$

which is equivalent to $\tan\varphi = -\kappa/\sinh\Theta$ but makes the two roles transparent: η sets the geometric prefactor, while Θ carries the chart singularity through $\coth\Theta$.

Direct dependence on the mixing angle θ. Substituting the exact influence-ratio definition $\tan\eta = -\cot\theta\,(\Sigma_\perp/\Sigma_z)$ from Eq. (E13) directly into Eq. (E40) gives

$$\tan\varphi = \cot\theta\,\frac{\Sigma_\perp}{\Sigma_z}\,(\cosh a + \sinh a\,\coth\Theta), \qquad a = \frac{\beta\omega_q}{2}. \quad \text{(E41)}$$

This is the explicit form in which the dependence of the *tilt angle* on the *mixing angle* is visible at a glance.

Temperature-dependent sign of the transverse/longitudinal ratio. Equation (E41) also makes clear that the sign of the tilt-angle chart variable is controlled by the product

$$X = \cot\theta\,\frac{\Sigma_\perp}{\Sigma_z}\,B(\Theta, a), \qquad B(\Theta, a) \equiv \cosh a + \sinh a\,\coth\Theta, \quad \text{(E42)}$$

so that

$$\operatorname{sgn} X = \operatorname{sgn}(\cot\theta)\,\operatorname{sgn}\left(\frac{\Sigma_\perp}{\Sigma_z}\right)\operatorname{sgn} B(\Theta, a). \quad \text{(E43)}$$

This isolates a second source of apparent angular "flips" besides the principal-arctan branch cut: the ratio $\Sigma_\perp/\Sigma_z$ itself can change sign as a function of temperature.

In the rewritten channel coefficients, the transverse coefficient $\Sigma_\perp(\beta)$ is a difference of contributions (with different temperature dependence), so it can pass through zero while $\Sigma_z(\beta)$ retains its sign on the same parameter slice. When this happens,

$$\Sigma_\perp(\beta_0) = 0 \implies \tan\eta(\beta_0) = 0 \implies m_\perp(\beta_0) = 0, \quad \text{(E44)}$$

and for temperatures on opposite sides of β_0 one has opposite signs of $\tan\eta$ (equivalently opposite signs of the transverse tilt prefactor in Eq. (E41)).

This is distinct from the branch locus $\Theta = 0$: a sign flip of $\Sigma_\perp/\Sigma_z$ changes the *orientation side* of the tilt angle, whereas $\Theta = 0$ drives the *principal-chart singularity.* In particular, the branch scale χ_b from Eq. (E20) depends on $\cos\eta$ and is therefore insensitive to $\eta \to -\eta$ (for a fixed sign of Δ_z), while the sign of $\tan\eta$ still flips. This is an additional reason the angle must be continued by continuity: otherwise a temperature-driven sign change

of $\Sigma_\perp/\Sigma_z$ can be misread as a geometric reflection rather than a smooth passage through $m_\perp = 0$.

Using the trigonometric identity

$$\cot\theta = \tan\Big(\frac{\pi}{2} - \theta\Big)\,, \tag{E45}$$

one may equivalently read the geometric prefactor as a $\pi/2$-*shifted* tangent. This is the clean way to see why the branch-corrected continuation of the tilt angle can look like a precession "around" the apparent reflected branch in a principal-angle plot.

Explicit vs implicit θ-dependence. Equation (E41) separates two effects:

1. **Explicit geometric factor:**

$$\cot\theta, \tag{E46}$$

which directly amplifies/suppresses the tilt as the coupling geometry is moved toward longitudinal/transverse alignment.

2. **Implicit dressing dependence through $\Theta(\beta, g, \theta)$:**

$$\Theta(\beta, g, \theta) = \operatorname{arctanh}\big(u(\beta, g, \theta)\big) - a, \qquad u = \gamma(\chi) s^2 \Sigma_z, \tag{E47}$$

with

$$\chi = g^2 s\sqrt{c^2\Sigma_\perp^2 + s^2\Sigma_z^2}, \qquad c = \cos\theta,\ s = \sin\theta. \tag{E48}$$

Thus θ also feeds back into the longitudinal rapidity through the crossover function $\gamma(\chi) = \tanh\chi/\chi$.

Local singular expansion (tilt-chart branch point). Near the branch scale χ_b (equivalently $\Theta = 0$), use $\coth\Theta = \Theta^{-1} + \Theta/3 + O(\Theta^3)$ to obtain

$$\tan\varphi = \cot\theta\,\frac{\Sigma_\perp}{\Sigma_z}\left(\frac{\sinh a}{\Theta} + \cosh a + \frac{\sinh a}{3}\Theta + O(\Theta^3)\right). \tag{E49}$$

This makes the branch anatomy transparent: the divergence driving the principal-angle jump comes from the Θ^{-1} term, while the overall sign and amplitude of that divergence are modulated by the geometric prefactor $\cot\theta\,(\Sigma_\perp/\Sigma_z)$.

Take arctan*: principal branch plus continuity index.* Define

$$X(\chi; \beta, \theta) \equiv -\tan\eta\,(\cosh a + \sinh a\,\coth\Theta(\chi))\,, \tag{E50}$$

so that $\tan\varphi = X$. The principal chart gives

$$\varphi_{\rm pr} = \arctan X \in \Big(-\frac{\pi}{2}, \frac{\pi}{2}\Big)\,, \tag{E51}$$

but the continuous physical angle in the coupling plane is

$$\varphi_{\rm cont} = \arctan X + n(\chi)\pi, \qquad n(\chi) \in \mathbb{Z}, \tag{E52}$$

where $n(\chi)$ is fixed by continuity (equivalently by angle unwrapping). At $\chi = \chi_b$, one has $X \to \pm\infty$ and therefore

$$\varphi_{\rm pr} \to \pm\frac{\pi}{2}, \tag{E53}$$

so the apparent branch "flip" is exactly the principal arctan chart crossing its boundary.

Fig. 7(b): precession vs the apparent reflected branch. Panel (b) of Fig. 8 uses the folded horizontal coordinate $\sqrt{m_x}$, which suppresses transverse sign information. In this folded representation, a principal-angle branch crossing can appear as a reflection toward $\pi - \alpha_f$ (with α_f the coupling-axis direction in the same chart). The reason is simple: once the transverse sign is folded out, the chart no longer distinguishes an angle from its reflected image $\alpha \mapsto \pi - \alpha$. The correct continuation is instead

$$\varphi_{\rm cont} = \varphi_{\rm pr} + n\pi, \tag{E54}$$

which carries the angle *through* the $\pm\pi/2$ chart boundary and yields a continuous precession around the Bloch disk. In that continuation, the trajectory is read as continuing toward alignment with the coupling direction $\hat{\mathbf{f}}$ (modulo the expected sign conventions and any director-vs-vector identification), rather than physically reflecting at the principal-chart image $\pi - \alpha_f$.

It is useful to state this in terms of a relative angle to the interaction axis. Let

$$\delta_f \equiv \varphi - \alpha_f. \tag{E55}$$

In the principal chart one sees

$$\delta_{f,\rm pr} = \varphi_{\rm pr} - \alpha_f, \tag{E56}$$

which can jump when $\varphi_{\rm pr}$ hits $\pm\pi/2$. The continuity-corrected relative angle is instead

$$\delta_{f,\rm cont} = \varphi_{\rm cont} - \alpha_f = \delta_{f,\rm pr} + n\pi, \tag{E57}$$

with the same branch index $n(\chi)$ as in Eq. (E52). The apparent "reflection" at $\pi - \alpha_f$ is therefore the principal-chart image of a trajectory whose continuous continuation is still precessing in the same sense. In particular, the correct interpretation of an apparent approach to $\pi - \alpha_f$ in the folded chart is a branch-resolved continuation of δ_f toward

$$\delta_f \to 0 \quad \text{(vector angle)}, \qquad \text{or} \qquad \delta_f \to 0 \pmod{\pi} \quad \text{(director/axi} \tag{E58}$$

i.e. alignment with $\hat{\mathbf{f}}$ (or its axis), not a physical bounce off the interaction axis.

By contrast, the Cartesian state components are smooth. In terms of the signed coupling-plane components, a branch-safe orientation angle is obtained from

$$\phi_{\rm wrap} \equiv \operatorname{atan2}\Big(m_\perp^{(\rm signed)}, -m_z\Big)\,, \qquad \phi_{\rm cont} \equiv \operatorname{unwrap}(\phi_{\rm wrap})\,, \tag{E59}$$

where "unwrap" denotes the standard numerical branch-tracking procedure that adds/subtracts integer multiples of 2π to enforce continuity.

Branch locus. The branch event is controlled by

$$\Theta = 0 \quad \Longleftrightarrow \quad u = \tanh a \quad \Longleftrightarrow \quad m_z = 0, \tag{E60}$$

using Eq. (E26). Equivalently, this is the condition $\chi = \chi_b$ from Eq. (E19). At this locus the principal angle inferred from Eq. (E40) can jump branches even though the exact Bloch vector remains continuous.

What changes and what does not. Equation (E41) shows the separation cleanly:

- changing the mixing angle θ changes the *prefactor* $\tan\eta$ (and therefore the tilt amplitude/sign),
- crossing $\Theta = 0$ changes the *chart branch* of the principal inverse tangent through the $\coth\Theta$ singularity.

This is precisely why the branch discussion should be phrased in terms of the tilt angle φ, not as a branch of the mixing angle θ.

Continuity enforcement for the tilt angle. If one treats φ as a *vector* angle, continuity is imposed by tracking ϕ_{cont} in Eq. (E59) (or equivalently $\varphi_{\text{cont}} = \arctan X + n\pi$). If one instead treats the orientation as an *axis* (director) where antipodal angles are equivalent modulo π, the local continuation across the principal-branch cut is implemented by the reflected chart

$$\varphi \mapsto \pi - \varphi \qquad \text{(axis/director continuation)}. \tag{E61}$$

This is the relevant reflection operation for the tilt-angle branch discussion: it is a *chart continuation rule*, not a change of the fixed coupling parameter θ in the Hamiltonian itself. In the folded Fig. 7(b) geometry, this is precisely the rule that turns an apparent reflection into a continuous precession toward the interaction axis.

Physical vs chart reorientation. The sign of m_z is a physical observable and is controlled by Eq. (E26), i.e. by the sign of $u - \tanh a$ (the denominator stays positive by Eq. (E30)). A branch choice for φ cannot by itself produce a physical m_z sign flip; it can only change the angle chart used to represent the same Cartesian Bloch vector.

Plotting-folding remark. The production Bloch-disk portrait uses the horizontal coordinate $\sqrt{m_x} = \sqrt{\langle\sigma_x\rangle}$, which folds the sign of the transverse component. In the manuscript coupling-plane gauge this can obscure the sign structure of $m_\perp^{(\text{signed})}$ and make a branch event appear as a geometric reflection.

7. Practical criterion: when does the branch flip track the crossover?

The crossover variable is χ (with the standard reference scale at $\chi = 1$), whereas the branch event is at $\chi = \chi_b$ from Eq. (E20). The clean comparison is therefore:

$$\chi_b < 1, \quad \chi_b = 1, \quad \chi_b > 1. \tag{E62}$$

In a coupling sweep at fixed temperature (or along a temperature sweep at fixed coupling), the apparent tilt-angle "flip" occurs when the trajectory crosses the branch locus $\chi = \chi_b$. If χ_b lies close to the crossover scale $\chi = 1$, the visual branch change and the crossover markers in Fig. 8(b) will appear to coincide. This is the precise sense in which the crossover can *cause* the observed branch flip: not because $\chi = 1$ is the branch condition identically, but because the physical path crosses the two nearby loci in the same region of parameter space.

Consequence for the purity argument. The exact purity statement is unchanged by any branch choice:

$$r = \sqrt{m_z^2 + m_\perp^2} = \tanh\chi, \tag{E63}$$

so the crossover interpretation based on χ remains valid. In particular, the branch event ($\chi = \chi_b$), the sign change of $\Sigma_\perp/\Sigma_z$, and the principal-angle jump in φ_{pr} do *not* alter the exact radius r of the Bloch vector.

What *is* affected is any geometric purity inference made from a folded projection plus a principal-angle chart. In panel (b), the horizontal folding $\sqrt{m_x}$ and a discontinuous principal φ_{pr} can make a continuous trajectory look as if it "reflects" and changes branch abruptly. If that reflected geometry is then used to reason about the approach to the ultrastrong limit, one can incorrectly attribute an orientation-chart artifact to the purity crossover itself. The correct purity reading is therefore:

- use χ (or equivalently $r = \tanh\chi$) for the radius/purity crossover,
- use a continuity-corrected angle (φ_{cont} or 'atan2'+unwrap) for the orientation.

With this separation, a branch-corrected continuation can precess toward $\hat{\mathbf{f}}$ while the purity still evolves monotonically through the same crossover region. The apparent coincidence in Fig. 7(b) is then interpreted as proximity of loci in parameter space (e.g. $\chi_b \approx 1$), not as a failure of the purity argument itself.

The operational rule for interpreting panel (b) is therefore:

> Track the tilt angle with the continuity-corrected branch $\varphi_{\text{cont}} = \arctan X + n\pi$ (or an equivalent 'atan2'+unwrap construction), and interpret the folded $\sqrt{m_x}$ plot as a projection of a continuous precession. The apparent reflection at the principal-branch image $\pi - \alpha_f$ is a chart artifact of the folded representation. With the correct continuation, the trajectory is read as precessing through the branch cut and toward alignment with $\hat{\mathbf{f}}$ (subject to the manuscript's vector/director sign conventions).

8. Standalone diagnostic plots supporting the branch-continuation interpretation

Figures 10, 11, and 12, together with 13, and 14, are the standalone diagnostics generated for this appendix from the exact qubit formulas. They are included here so the branch-continuation argument can be checked directly against the plotted magnetisation trajectories and branch-sheet constructions, rather than inferred only from the folded geometry in Fig. 8(b).

Appendix F: RG-Like Branch Bifurcation and Involutive Crossover in the Qubit Mean-Force Geometry

Appendix E develops the branch bookkeeping in detail. Here we recast the same logic in the form that is most useful for interpretation: the purity is governed by a single running scalar χ, while the apparent multiplicity resides in the continuation of the angle chart. The resulting picture is naturally phrased in RG-like language, but with an important caveat stated at the end of this section: for every finite (β, g) the exact equilibrium state is unique.

Running coupling, purity, and the involutive line. For the qubit geometry of Sec. and Sec. , the exact purity is controlled by

$$\chi(\beta, g) = g^2 \chi_0(\beta), \tag{F1}$$

so that

$$r(\beta, g) = \tanh \chi(\beta, g). \tag{F2}$$

This is the sense in which χ plays the role of a running coupling (equivalently, a purity-flow variable).

The crossover $\chi = 1$ defines mutually inverse maps on the monotone branch of $\chi_0(\beta)$ used throughout the figures:

$$g_\star(\beta) = \chi_0(\beta)^{-1/2}, \qquad \beta_\star(g) = \chi_0^{-1}(g^{-2}), \tag{F3}$$

or, equivalently,

$$\chi = 1 \iff g = g_\star(\beta) \iff \beta = \beta_\star(g). \tag{F4}$$

The line $\chi = 1$ is therefore self-dual (involutive) under the exchange between the two parameterizations of the same crossover.

The formal singularity of the principal tilt-angle chart remains $\Theta = 0$ (Appendix E). What follows does not replace that statement. Rather, it identifies $\chi = 1$ as the geometrically natural gluing locus for a branch-continuation construction that resolves the apparent reflection visible in the folded representation of Fig. 8.

A full-Bloch-circle view of the branch split. Figure 15 removes the $\sqrt{m_x}$ fold and plots the same geometry in the full coupling-plane Bloch coordinates $(m_\perp^{(\mathrm{signed})}, m_z)$. For a representative coupling $g > g_\star^{(\mathrm{ref})}$, a single incoming branch is drawn on the high-temperature side and glued at $T_\star(g)$ (equivalently, $\chi = 1$) to the ray $\pi - \theta$. Below $T_\star(g)$, two continuations are shown: one returns toward the ground-state direction, while the other is the z-reflected continuation toward the θ sector (equivalently, a sign-flipped branch image in the plotted transverse coordinate). This is exactly the bifurcation topology one wants to see in the unfolded magnetisation plane.

The key point is that the radius is never altered in this construction. The continuation acts only on the angular sheet, while the exact purity remains $r = \tanh \chi$.

The RG-like bifurcation variable. The relative scaling of thermal and coupling flow is captured by the single coordinate

$$y \equiv \log \chi = \log\bigl(g^2 \chi_0(\beta)\bigr) = 2 \log\bigl(g/g_\star(\beta)\bigr), \tag{F5}$$

for which the involutive line is simply $y = 0$.

In this variable the geometry becomes especially transparent: one branch sheet approaches the hub at $y = 0$, and for $y > 0$ the continuation can be taken to different asymptotic angular sectors. In the representative construction of Fig. 16, the two displayed low-temperature continuations approach the ground-state ray (on the lifted 2π sheet) and the θ sector, respectively. The purity panel then seals the interpretation: the apparent multiplicity is angular, not radial.

Where this lives in the (T, g) plane. Figure 17 plots the same structure in the parameter plane using the color field $y = \log \chi$ and the contour $\chi = 1$. The contour is drawn both directly from $g_\star(\beta)$ and indirectly from sampled inverse points $(\beta_\star(g), g)$; their coincidence is the numerical expression of the involutive relation on the monotone branch of $\chi_0(\beta)$ used here. The companion panel presents a representative RG flow cut in the same language, comparing $\log_{10} \chi_0(\beta)$ to $\log_{10} \chi(\beta; g_{\mathrm{ref}})$ for a fixed $g_{\mathrm{ref}} > g_\star^{(\mathrm{ref})}$, so that the crossover appears simply as the zero crossing of the latter curve.

For the displayed domain, the sampled grid remains in the $\Theta < 0$ sector, so the formal principal-chart singularity $\Theta = 0$ does not enter the plot. This is precisely why it is useful to separate the two statements: $\Theta = 0$ is the exact singular locus of the principal tilt-angle chart, whereas $\chi = 1$ is the self-dual crossover line that organizes the continuation geometry shown here.

Interpretation. The figures support an RG-like reading in which the crossover line $\chi = 1$ acts as a branch-gluing threshold, and the low-temperature continuation can be organized into distinct asymptotic angular-sector attractors (fixed directions on the branch sheets). The exact finite-parameter equilibrium state, however, remains single-valued. What is multivalued here is the con-

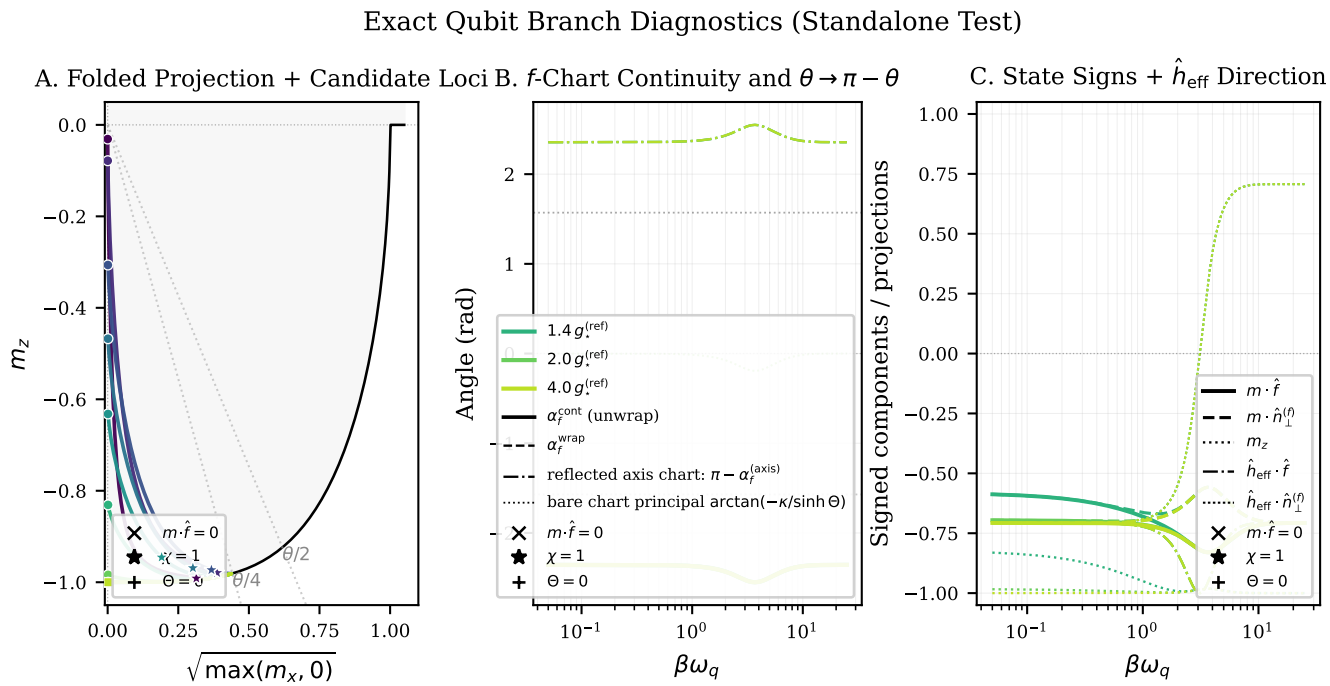

FIG. 10. Standalone branch-diagnostics overview (exact qubit model). Panel A reproduces the folded-projection geometry relevant to Fig. 8(b), while Panels B–C compare principal-chart and continuity-corrected angular descriptions together with signed component projections. The purpose of this figure is to expose how a folded/projection chart can make a continuous precession appear as a reflection, and to separate crossover markers ($\chi = 1$) from genuine angle-chart singular loci.

tinuation structure of the angle chart, not the equilibrium density matrix itself.

Appendix G: Tilt-Angle Branch Structure and MFG-Directed Continuations

The folded representation in Fig. 8(b) compresses the transverse sign and therefore hides part of the qubit geometry. The result is a visually suggestive "reflection" that is easy to misread as a dynamical reversal. The physically cleaner interpretation is a two-part statement:

1. the reduced-state tilt angle is a chart variable extracted through an inverse tangent and is therefore branch-valued;

2. the sign of the tilt-angle argument itself depends on a competition between thermal and coupling-driven scales.

Once these are separated, the branch family that aligns with the Cresser–Anders ultrastrong mean-force Gibbs (MFG) endpoint is the continuation that lands on the coupling-direction sector, not the branch that returns to the bare ground-state ray.

Exact geometry and what is actually singular. In the rewritten qubit parameterization (Sec.) the reduced-state magnetisation in the $(\hat{\mathbf{e}}_\perp^{(s)}, \hat{\mathbf{n}}_s)$ plane is

$$m_z = \tanh\Theta, \tag{G1}$$

$$m_\perp = \tan\eta\,(m_z \cosh a + \sinh a)\,, \qquad a \equiv \beta\omega_q/2, \tag{G2}$$

with

$$\tan\eta = -\cot\theta\,\frac{\Sigma_\perp}{\Sigma_z}. \tag{G3}$$

The principal tilt-angle chart is defined by

$$\tan\varphi = -\frac{m_\perp}{m_z} = -\tan\eta\,(\cosh a + \sinh a\,\coth\Theta)\,. \tag{G4}$$

The Cartesian Bloch vector $(m_\perp, m_z)$ is the physical object. The singularity is in the *chart* variable φ (through the inverse tangent), not in the state itself.

A rational form for the tilt-angle argument. Define

$$\tan\varphi = X(\beta, g, \theta), \qquad X \equiv \cot\theta\,\frac{\Sigma_\perp}{\Sigma_z}\,(\cosh a + \sinh a\,\coth\Theta)\,. \tag{G5}$$

With $s \equiv \sin\theta$ and

$$u \equiv \gamma s^2 \Sigma_z, \qquad t_a \equiv \tanh a, \qquad \Theta = \operatorname{arctanh}(u) - a, \tag{G6}$$

one has

$$\tanh\Theta = \frac{u - t_a}{1 - u t_a} \quad\Longrightarrow\quad \coth\Theta = \frac{1 - u t_a}{u - t_a}, \tag{G7}$$

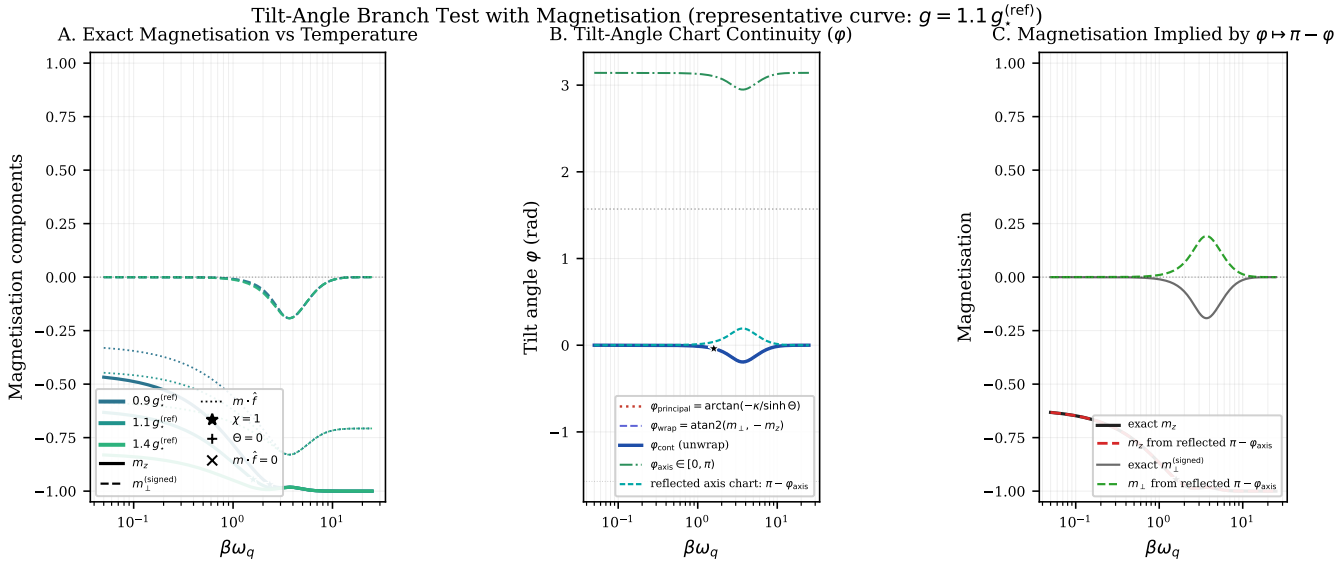

FIG. 11. Standalone tilt-angle and magnetisation test for the branch-continuation argument. The plotted magnetisation components make explicit that the physical Bloch vector remains continuous, while the principal tilt angle can jump by a chart branch crossing. Comparing the principal angle, continuity-corrected angle, and the reflected axis-chart continuation demonstrates the appendix claim: the visually apparent reflection in the folded representation is resolved by a continuous angular continuation toward alignment with $\hat{\mathbf{f}}$, not by a physical bounce at the principal-chart image $\pi - \alpha_f$.

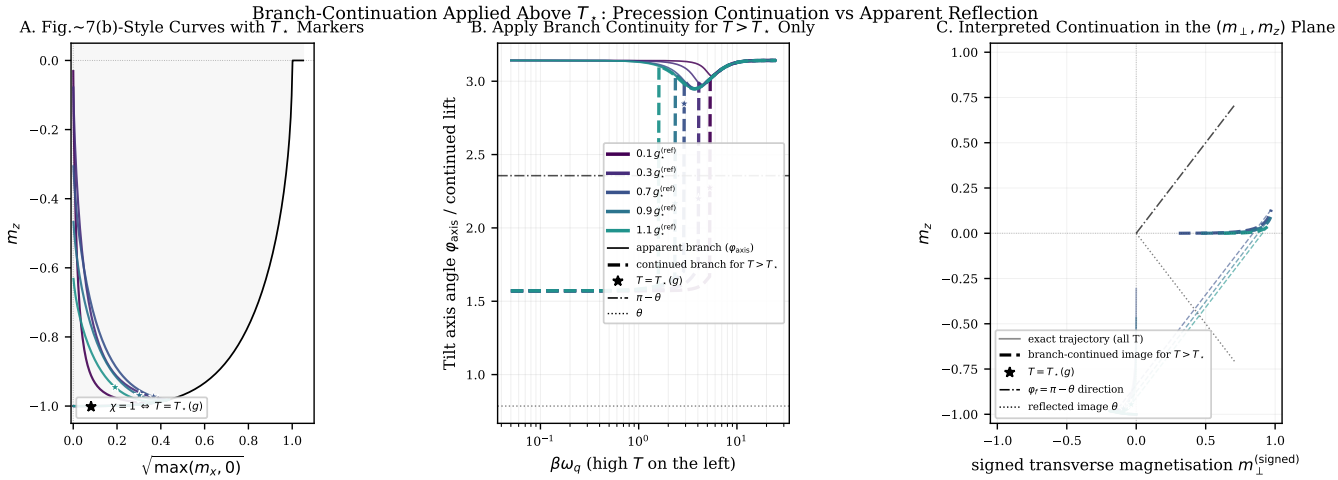

FIG. 12. Explicit branch-continuation construction applied to the Fig. 7(b)-style curves. The crossover marker $T = T_\star(g)$ (equivalently $\chi = 1$) partitions each trajectory, and the plotted overlay applies the director/axis continuity rule only on the $T > T_\star$ segment (high-temperature side) so that the tilt-angle branch is continued as a precession rather than read as a reflection at the principal-chart image. Panel B shows the piecewise continuation in the tilt-axis angle chart relative to the reference directions θ and $\pi - \theta$; Panel C maps that continuation back into the $(m_\perp, m_z)$ plane as an interpretive branch-resolved overlay. This figure is the direct visualization of the appendix claim that the apparent reflection can be re-read as a continuity-corrected approach toward the interaction-axis sector.

and therefore

$$\cosh a + \sinh a \, \coth\Theta = \frac{u \operatorname{sech} a}{u - t_a}. \tag{G8}$$

Substituting this into Eq. (G5) yields the exact rationalized form

$$X = \cot\theta \, \frac{\Sigma_\perp}{\Sigma_z} \, \frac{u \operatorname{sech} a}{u - t_a} = \cot\theta \, \gamma s^2 \Sigma_\perp \, \frac{\operatorname{sech} a}{u - t_a}. \tag{G9}$$

This is the key simplification. The numerator carries the transverse sign through $\Sigma_\perp$, while the denominator

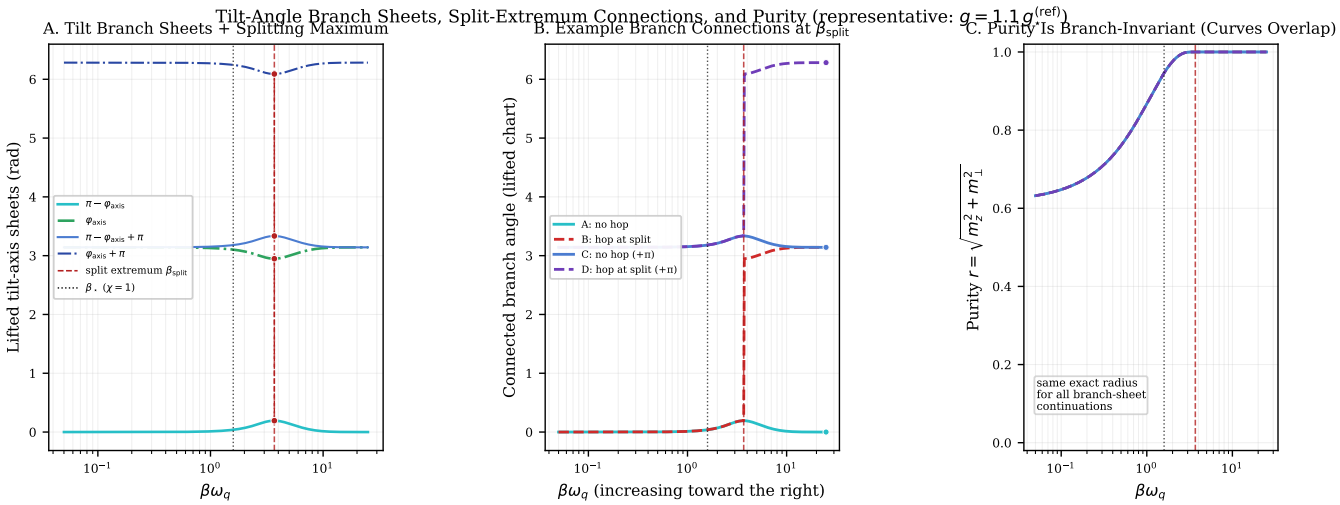

FIG. 13. Tilt-angle branch-sheet construction (representative coupling) showing how different lifted branch connections can be spliced at the splitting extremum of the visible branch separation, and how those distinct branch continuations flow to different asymptotic angle values as $\beta \to \infty$. Panel A displays the lifted tilt-axis sheets and marks both the crossover point $\beta_\star$ ($\chi = 1$) and the splitting extremum β_{split}. Panel B shows example branch connections obtained by switching sheets at β_{split}. Panel C plots the corresponding purities and shows that they are branch-invariant: all branch-sheet continuations share the same exact radius $r = \sqrt{m_z^2 + m_\perp^2} = \tanh\chi$. This is the visual "seal" of the appendix argument that the branch multiplicity is an angular-sheet/continuation structure on top of a single purity crossover.

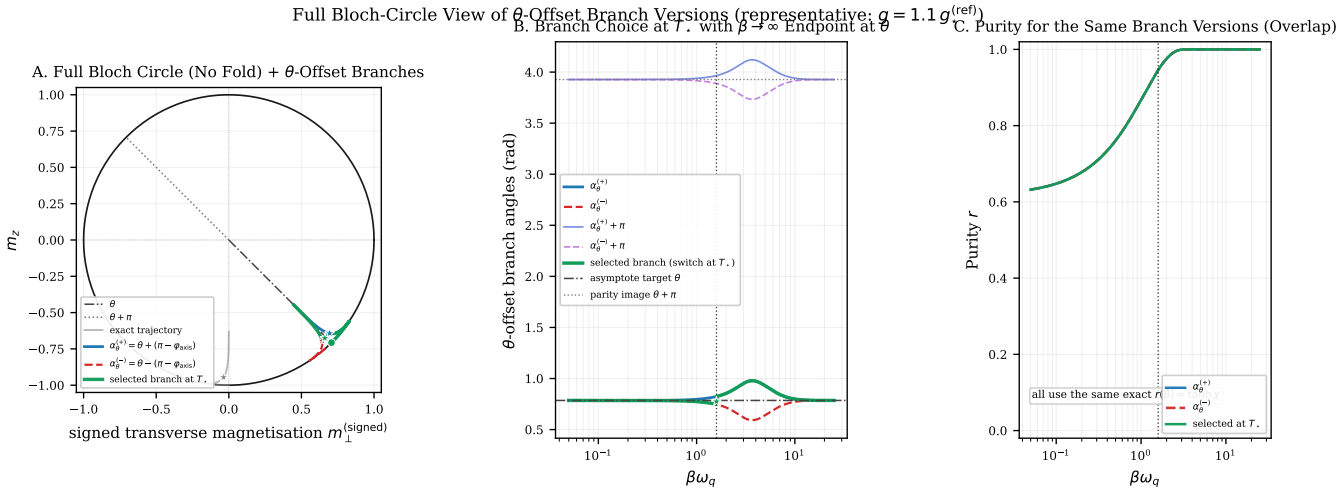

FIG. 14. Full Bloch-circle (unfolded) visualization of θ-offset branch versions for a representative coupling. Unlike the folded $\sqrt{m_x}$ portrait, Panel A shows the full coupling-plane circle and overlays branch versions constructed so that one selected branch (green) switches at $T_\star$ and approaches the original angle θ as $\beta \to \infty$. Panel B shows the corresponding θ-offset angle versions and the explicit branch choice at $T_\star$, together with the asymptote targets θ and $\theta + \pi$. Panel C confirms that all these branch versions share the same exact purity $r(\beta) = \tanh\chi$. This figure is the appendix's direct answer to the "full Bloch circle" branch-continuation question: the branch choice changes the angular sheet/endpoint interpretation while leaving the exact purity unchanged.

carries the thermal–coupling competition through

$$u(\beta, g) - \tanh a(\beta). \tag{G10}$$

Thus the sign of the tilt-angle argument is controlled by the *relative scaling* of β and g:

$$\operatorname{sgn} X = \operatorname{sgn}(\cot\theta)\operatorname{sgn}(\Sigma_\perp)\operatorname{sgn}\big(u - \tanh a\big). \tag{G11}$$

The chart enhancement occurs when $u \approx \tanh a$ (equivalently $m_z \approx 0$), but sign changes can also be triggered by $\Sigma_\perp$ changing sign. These are distinct mechanisms.

Continuous angle branches and the geometric hub. The continuous vector-angle branch is

$$\varphi_{\mathrm{cont}} = \arctan X + n\pi, \qquad n \in \mathbb{Z}, \tag{G12}$$

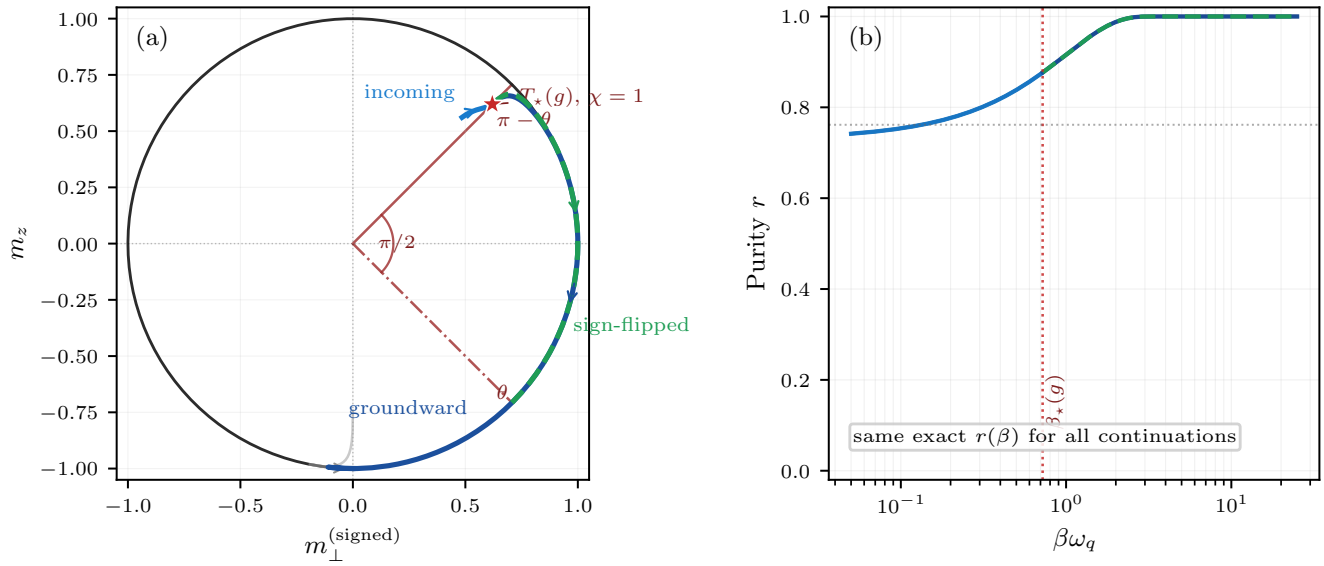

FIG. 15. Full Bloch-circle branch-continuation picture for a representative $g = 1.25\, g_\star^{(\mathrm{ref})}$. In panel (a), the incoming branch is glued at $T_\star(g)$ (red star, $\chi = 1$) to the $\pi - \theta$ ray, and two post-$T_\star$ continuations are shown: a ground-state-sector branch and a z-reflected branch toward the θ sector. The pale grey curve is the exact equilibrium trajectory, included only as geometric context. Panel (b) shows the corresponding purity: all displayed continuations lie on the same exact curve $r = \sqrt{m_z^2 + m_\perp^2} = \tanh \chi$.

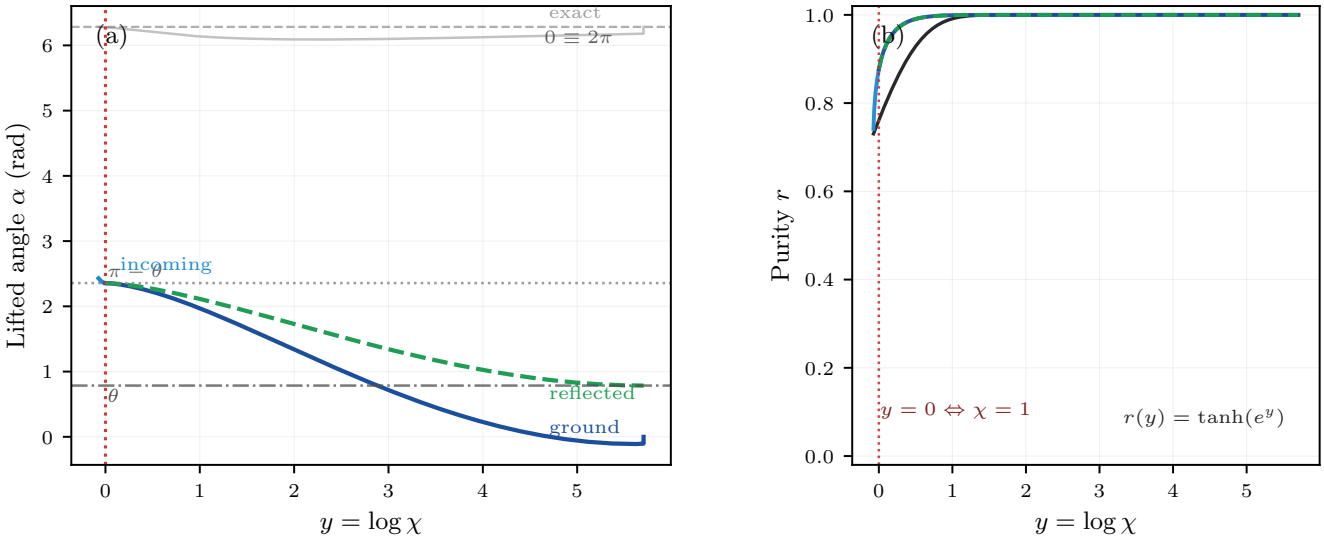

FIG. 16. RG-like branch bifurcation in the single flow coordinate $y = \log \chi$. Panel (a) shows the branch sheets meeting at the involutive crossover $y = 0$ (equivalently $\chi = 1$) and separating into distinct asymptotic angular sectors for $y > 0$. Panel (b) shows the branch-invariant purity collapse onto the exact law $r(y) = \tanh(e^y)$.

while the axis-angle chart identifies φ and $\varphi + \pi$. For the unfolded Bloch-circle plots it is useful to define the axis angle

$$\varphi_{\mathrm{ax}} \equiv \varphi \pmod{\pi}. \tag{G13}$$

In the plotting convention used here (angle measured from the bare ground-state ray $-\hat{\mathbf{n}}_s$ toward the positive transverse direction), the coupling-axis ray is $\pi - \theta$. The visible branch-selection geometry is therefore organized by the *hub condition*

$$\varphi_{\mathrm{ax}} = \pi - \theta \pmod{\pi}, \tag{G14}$$

which need not coincide with the principal-chart singularity $u = \tanh a$.

Near the hub, the local branch family is naturally two-

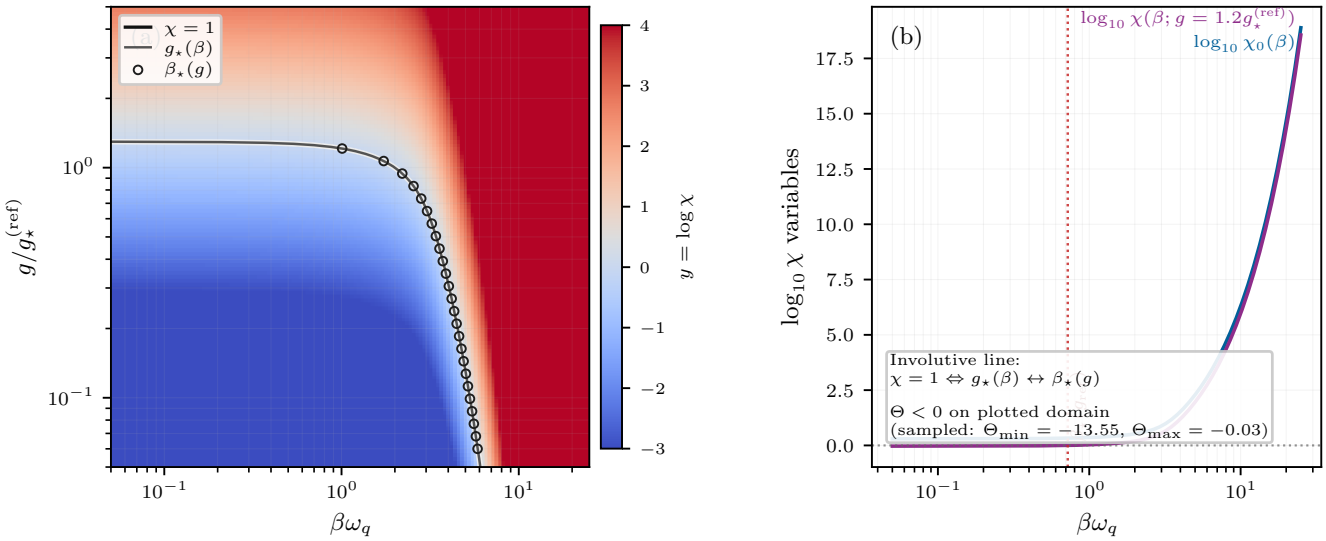

FIG. 17. Involutive crossover geometry generated by $\chi(\beta, g) = g^2\chi_0(\beta)$. Panel (a) shows the field $y = \log\chi$ and the contour $\chi = 1$, overlaid with both constructions of the same crossover line: $g_\star(\beta)$ and sampled inverse points $(\beta_\star(g), g)$. Panel (b) shows a representative RG flow cut in $\log_{10}\chi$ variables: the monotone seed $\log_{10}\chi_0(\beta)$ and the driven flow $\log_{10}\chi(\beta; g_{\mathrm{ref}})$, with the vertical marker at $\beta_\star(g_{\mathrm{ref}})$. The inset text reports the sampled Θ range, which remains negative on the plotted domain.

valued on the axis chart,

$$\varphi_{\mathrm{ax}}^{(+)} \sim \pi - \theta, \qquad \varphi_{\mathrm{ax}}^{(-)} \sim \theta, \tag{G15}$$

and four-valued on a lifted vector chart through parity shifts,

$$\varphi_{\mathrm{cont}}^{(\pm)} + \pi. \tag{G16}$$

This is the source of the "two or four branches" language: a single geometric hub together with sign multiplicity and inverse-tangent sheet choice.

Which branch is physically relevant for MFG alignment? The ultrastrong MFG state quoted by Cresser and Anders for a qubit coupled along $\hat{\mathbf{r}}$ is

$$\rho_S^{(\lambda\to\infty)} = \frac{1}{2}\Big[\mathbf{1} - \sigma_{\hat{\mathbf{r}}}\tanh\big(\tfrac{1}{2}\beta\omega_q\cos\theta\big)\Big]. \tag{G17}$$

In our notation $\hat{\mathbf{r}} \equiv \hat{\mathbf{f}}$. The MFG-compatible branch is therefore the continuation whose endpoint lies on the *coupling-direction sector* (the θ ray in the plotting convention), not the continuation that returns to the bare ground-state ray.

This can be made explicit by comparing the exact equilibrium ultrastrong endpoint to the MFG target. From Appendix E,

$$\Theta(\chi) = \operatorname{arctanh}\big(\cos\eta\,\tanh\chi\big) - a, \qquad a = \beta\omega_q/2, \tag{G18}$$

so at fixed (β, θ),

$$\Theta_\infty = \lim_{g\to\infty}\Theta = \operatorname{arctanh}(\cos\eta) - a, \tag{G19}$$

because $\chi = g^2\chi_0 \to \infty$ implies $\tanh\chi \to 1$. Substituting into Eq. (G2) gives the exact equilibrium ultrastrong endpoint

$$\mathbf{m}_\infty^{(\mathrm{ex})} = \frac{\sin\eta\,\hat{\mathbf{e}}_\perp^{(s)} + \big(\cos\eta\cosh a - \sinh a\big)\hat{\mathbf{n}}_s}{\cosh a - \cos\eta\,\sinh a}, \tag{G20}$$

with exact purity

$$r_\infty^{(\mathrm{ex})} = \lim_{g\to\infty}\tanh\chi = 1. \tag{G21}$$

By contrast, the MFG endpoint in our notation is

$$\rho_{S,\infty}^{(\mathrm{MFG})} = \frac{1}{2}\big[\mathbf{1} - \sigma_{\hat{\mathbf{f}}}\,r_{\mathrm{MFG}}\big], \qquad r_{\mathrm{MFG}} \equiv \tanh(a\cos\theta), \tag{G22}$$

so MFG consistency requires both

$$\varphi_{\mathrm{ax}}^{(\mathrm{br})}(g\to\infty) = \theta \pmod{\pi}, \tag{G23}$$

$$r_{\mathrm{br}}(g\to\infty) = \tanh(a\cos\theta). \tag{G24}$$

Equation (G21) then shows the obstruction directly: a pure chart relabeling can match the *direction*, but not the finite-temperature MFG *radius*.

Correlated-limit viewpoint and branch selection. The physically persuasive way to state the MFG comparison is to keep the effective thermal variable

$$a_{\mathrm{eff}} \equiv \frac{\beta\omega_q}{2}\cos\theta \tag{G25}$$

visible in the asymptotic endpoint. Then the MFG radius is simply

$$r_{\mathrm{MFG}} = \tanh a_{\mathrm{eff}}, \tag{G26}$$

with density-matrix purity

$$\mathcal{P}_{\mathrm{MFG}} = \frac{1}{2}\big(1 + r_{\mathrm{MFG}}^2\big). \tag{G27}$$

In this correlated-limit language the endpoint radius is fixed by a_{eff}, and the remaining freedom is angular. The two post-hub continuations therefore correspond to distinct endpoint directions:

$$\mathbf{m}_{\mathrm{br}}^{(\mathrm{ground})} \to -\tanh(a_{\mathrm{eff}})\,\hat{\mathbf{n}}_s, \tag{G28}$$

$$\mathbf{m}_{\mathrm{br}}^{(\mathrm{MFG})} \to -\tanh(a_{\mathrm{eff}})\,\hat{\mathbf{f}}. \tag{G29}$$

The second is the Cresser–Anders-compatible branch. In other words: once the endpoint is required to retain a finite thermal imprint through a_{eff}, the continuation that runs back to the bare ground state is no longer the physically relevant branch for MFG matching.

A simple branch-family ansatz implementing this idea is

$$\mathbf{m}_{\mathrm{br}}(\beta) = r_{\mathrm{MFG}}(\beta)\,\big[\sin\alpha_{\mathrm{br}}(\beta)\,\hat{\mathbf{e}}_\perp^{(s)} - \cos\alpha_{\mathrm{br}}(\beta)\,\hat{\mathbf{n}}_s\big], \tag{G30}$$

where $\alpha_{\mathrm{br}}(\beta)$ is selected from the local branch family so that the trajectory passes through the hub and asymptotes to the θ sector. This is a branch-switched continuation rule, not an exact equilibrium trajectory.

Purity diagnostics: exact law versus branch-switched families. For the exact equilibrium qubit state,

$$r_{\mathrm{exact}}(\beta, g) = \sqrt{m_z^2 + m_\perp^2} = \tanh\chi(\beta, g). \tag{G31}$$

If one only relabels the angle chart, the purity is unchanged. If one instead reconstructs a new branch-switched state trajectory, the purity becomes an independent diagnostic:

$$r_{\mathrm{br}}(\beta) \equiv \sqrt{\big[m_z^{(\mathrm{br})}(\beta)\big]^2 + \big[m_\perp^{(\mathrm{br})}(\beta)\big]^2}. \tag{G32}$$

This distinction is essential for interpretation: branch multiplicity lives in the angle chart, while MFG matching imposes an additional radial requirement.

Figure: what should be read from it. Figure 18 is designed to isolate the three ingredients that were entangled in the folded Fig. 8(b): the geometric hub at $\pi - \theta$, the post-hub branch choice, and the radial law. Panel (a) shows the unfolded Bloch-circle geometry and the two post-hub continuations. Panel (b) shows the same construction on the angle chart. Panel (c) shows the MFG-guided radius and purity used in the ansatz. The figure is intentionally presented as a branch-family construction rather than an exact equilibrium trajectory, because that is precisely the point of the MFG comparison.

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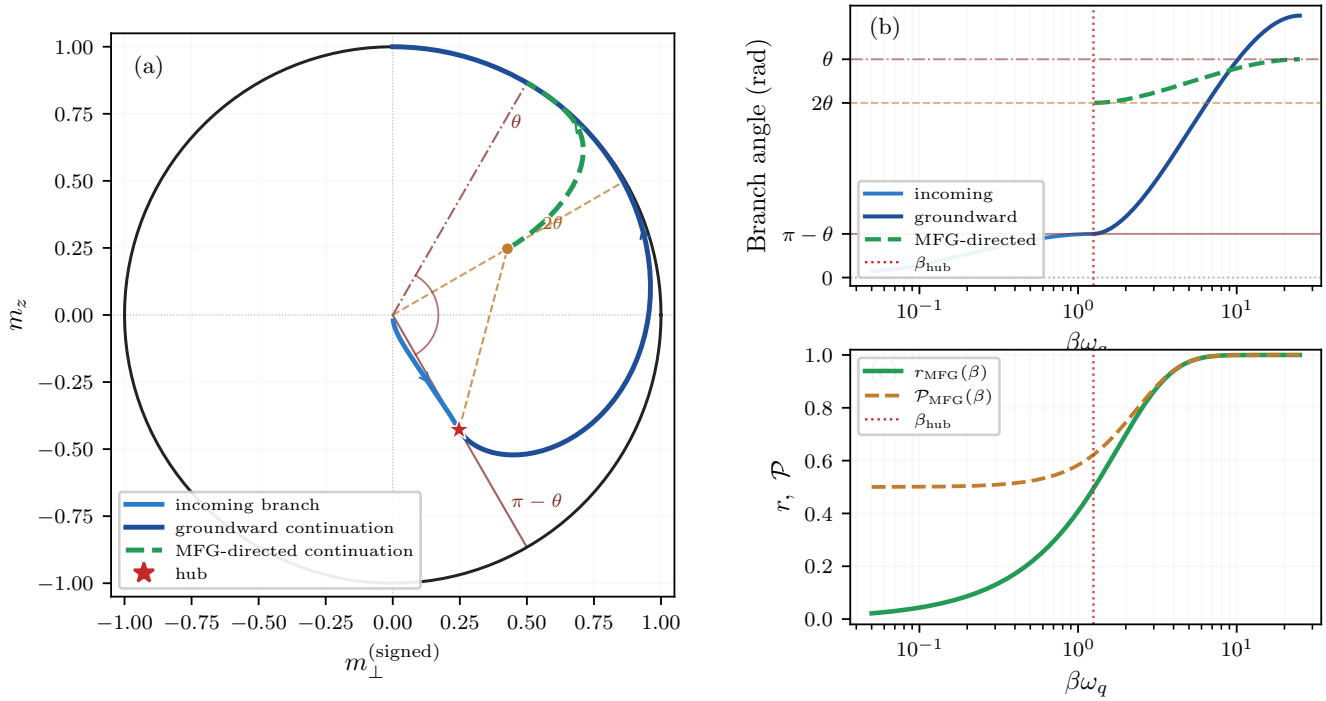

FIG. 18. Hub-centered branch family illustrating the tilt-angle interpretation and MFG-directed continuation (illustrative ansatz, not an exact equilibrium trajectory). (a) Unfolded Bloch-circle geometry in $(m_\perp^{(\mathrm{signed})}, m_z)$: an apparent branch reaches the hub $\varphi_{\mathrm{ax}} = \pi - \theta$, after which one continuation returns groundward while the alternate sign/arctan image (starting at the local 2θ point) precesses toward the coupling-direction sector θ. (b) The corresponding angle-branch construction. (c) The MFG-guided Bloch radius and density-matrix purity used in the branch-family ansatz, $r_{\mathrm{MFG}} = \tanh[(\beta\omega_q/2)\cos\theta]$ and $\mathcal{P}_{\mathrm{MFG}} = \frac{1}{2}(1 + r_{\mathrm{MFG}}^2)$.

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